

Gravity and Gravitation

5.1 Motion of Falling Bodies

It is a matter of common observation that bodies fall towards the ground. Leaves of the tree, fruits from the tree, a body just dropped from top of a tower, a stone thrown upward etc. all reach the ground. According to *Aristotle* it was believed that the time taken by a heavier body to reach the ground is less as compared to a lighter body dropped from the same height. This belief continued until the sixteenth century and no body either challenged it or tried to prove its correctness.

In 1590, Galileo disapproved this idea. He showed from his famous experiments at the leaning tower of Pisa that all bodies dropped from the same height reach the ground at the same instant irrespective of their masses.

Galileo showed that bodies of different masses when dropped from the top of the leaning tower of Pisa, reached the ground at the same time. A large crowd was present to see the correctness of Galileo's idea. When a stone and a piece of paper were dropped from the tower at the same time, it was found that the stone reached the ground earlier. This difference in time was explained by Galileo to be due to the resistance offered by air on the paper. He showed that if a paper and a stone are dropped from the same height in vacuum, both take the same time to fall through the height.

5.2 Kepler's Laws of Motion and Derivation of Law of Gravitation

The three laws of planetary motion are :

(1) **Shape of the orbit.** Every planet moves in an elliptical orbit with the sun being at one of its foci.

(2) **Velocity in the orbit [Law of areas].** The radius vector drawn from the sun to the planet sweeps out equal areas in equal intervals of time.

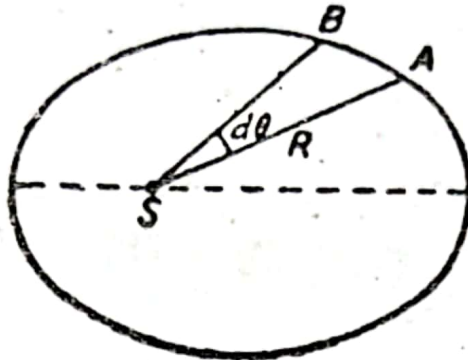


Fig. 5.1

Consider a planet moving in an elliptical orbit with the sun at the focus S (5.1). Let the radius vector of the planet describe a small angle $d\theta$ in time dt while moving from the position A to B . Arc $AB = R d\theta$.

Area swept in time dt

$$= \text{Area } ABS = \frac{1}{2} R \times R d\theta = \frac{1}{2} R^2 d\theta$$

$$\text{Areal velocity} = \frac{\frac{1}{2} R^2 d\theta}{dt} = \frac{1}{2} R^2 \left(\frac{d\theta}{dt} \right) = \frac{1}{2} R^2 \omega = \text{constant}$$

In this case $\frac{1}{2} R^2 \omega$ represents the areal velocity of the radius vector. When R decreases, ω increases but $\frac{1}{2} R^2 \omega$ is always constant, i.e., the areal velocity is constant.

As areal velocity $\frac{1}{2} R^2 \omega$ is constant, therefore $\frac{1}{2} M R^2 \omega$ is also a constant. Therefore $M R^2 \omega$ remains constant throughout the motion of a planet in its orbit. The term $M R^2 \omega$ represents the angular momentum of the planet. Hence the law of areas is a statement of the law of conservation of angular momentum.

$$\text{Area of the ellipse} = \pi ab$$

Here a and b are semi-major and semi-minor axes of the ellipse respectively. Time period of revolution of the planet around the sun,

$$T = \frac{\text{Area}}{\text{Areal velocity}}$$

$$T = \frac{\pi ab}{\frac{1}{2} R^2 \omega} = \frac{2\pi ab}{R^2 \omega}$$

(3) **Time periods of Planets.** The square of the time period of revolution of a planet around the sun is proportional to the cube of the semi-major axis of the orbit. Thus, if T_1 and T_2 are the semi-major axes of the planets respectively,

$$\frac{T_1^2}{a_1^3} = \frac{T_2^2}{a_2^3} = \text{constant}$$

or
$$\frac{T^2}{a^3} = \text{constant}$$

$$T^2 \propto a^3$$

5.3 Derivation of Law of Gravitation

Suppose the mass of a planet A is M_1 , the radius of its orbit is R_1 and time period of revolution is T_1 . It is assumed that the orbit is circular. The force of attraction exerted by the sun on the planet (centripetal force)

$$F_1 = M_1 R_1 \omega_1^2$$

$$F_1 = M_1 R_1 \left[\frac{2\pi}{T_1} \right]^2 \quad \dots (i)$$

Similarly for a second planet B of mass M_2 ; Radius R_2 and period of revolution round the sun T_2 ,

$$F_2 = M_2 R_2 \omega_2^2$$

or
$$F_2 = M_2 R_2 \left[\frac{2\pi}{T_2} \right]^2 \quad \dots (ii)$$

$$\frac{F_1}{F_2} = \left(\frac{M_1}{M_2} \right) \left(\frac{R_1}{R_2} \right) \left(\frac{T_2}{T_1} \right)^2 \quad (\dots iii)$$

But according to Kepler's third law,

$$\left(\frac{T_2}{T_1} \right)^2 = \left(\frac{R_2}{R_1} \right)^3 \quad \dots (iv)$$

Substituting these values in equation (iii)

$$\frac{F_1}{F_2} = \left(\frac{M_1}{M_2} \right) \left(\frac{R_2}{R_1} \right)^3$$

or
$$\frac{F_1 R_1^2}{M_1} = \frac{F_2 R_2^2}{M_2} = \text{const}$$

$$\therefore F \propto \frac{M}{R^2} \text{ or (i) } F \propto M \text{ and (ii) } F \propto \frac{1}{R^2}$$

Thus, the force of attraction exerted by the sun on a planet is proportional to its mass and inversely proportional to the square of its distance from the sun.

Newton concluded that the force which is exerted by the earth on the bodies on its surface is the same as the force under which the moon and the planets move. The force of attraction must be mutual i.e. a force exerted by a body on the earth must be equal and opposite to the force exerted by the earth on the body. Newton reduced the Kepler's laws into a single law of gravitation.

Example 5.1 (A). A satellite revolves round a planet in an elliptical orbit. Its maximum and minimum distances from the planet are 1.5×10^7 m and 0.5×10^7 m respectively. If the speed of the satellite

at the farthest point be 5×10^3 m/s, calculate the speed at the nearest point. [Lucknow, 1981]

According to Kepler's laws

$$\frac{1}{2} R^2 \omega = \text{const}$$

$$\therefore \frac{1}{2} R_1^2 \omega_1 = \frac{1}{2} R_2^2 \omega_2$$

or

$$\frac{1}{2} R_1 v_1 = \frac{1}{2} R_2 v_2$$

Here

$$R_1 = 1.5 \times 10^7 \text{ m}$$

$$R_2 = 0.5 \times 10^7 \text{ m}$$

$$v_1 = 5 \times 10^3 \text{ m/s}$$

$$v_2 = ?$$

$$v_2 = \left(\frac{R_1 v_1}{R_2} \right)$$

$$v_2 = \left(\frac{1.5 \times 10^7 \times 5 \times 10^3}{0.5 \times 10^7} \right)$$

$$v_2 = 1.5 \times 10^4 \text{ m/s.}$$

5.4 Newton's Universal Law of Gravitation

The motion of the planets round the sun led Newton to formulate the Universal Law of Gravitation. According to Newton, this law is true not only for the heavenly bodies but is also true for any two bodies in this universe. The law holds good for the force of attraction between the sun and the earth, for the attraction between the moon and the earth, for the attraction of an electron in the orbit and the nucleus in the atom etc.

According to this law :

Every body in this universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

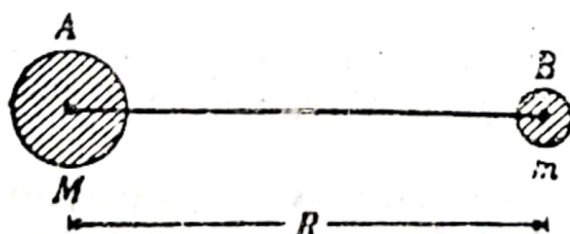


Fig. 5.22

Consider two bodies A and B of masses M and m respectively. The distance between their centres = R (Fig. 5.2)

$$F \propto Mm \quad \dots(i)$$

$$F \propto \frac{1}{R^2} \quad \dots(ii)$$

Combining (i) and (ii)

$$F \propto \frac{Mm}{R^2} \quad \text{or} \quad F = G \left[\frac{Mm}{R^2} \right]$$

Here G is the universal constant of gravitation

If $M = m = 1$,

and

$$R = 1,$$

$$F = G.$$

The constant of gravitation is numerically equal to the force of attraction between two unit masses at a unit distance apart.

In CGS units $G = 6.670 \times 10^{-8} \text{ dyne-cm}^2/\text{g}^2$

In rationalised MKS unit,

$$G = 6.670 \times 10^{-11} \text{ newton-m}^2/\text{kg}^2$$

Example 5.1 Assuming the earth's orbit around the sun to be a circle, calculate the angular velocity of the earth about the sun. What is its average linear speed and centripetal acceleration with respect to the sun? Orbital radius of earth = $1.5 \times 10^{11} \text{ m}$

Here $T = 365.25 \text{ days} = 365.25 \times 86400$

$$T = 3.156 \times 10^7 \text{ s}$$

(i) $\omega = \frac{2\pi}{T} = \frac{2\pi}{3.156 \times 10^7}$

$$\omega = 1.991 \times 10^{-7} \text{ radians/s}$$

(ii) $v = R\omega$

$$v = 1.5 \times 10^{11} \times 1.991 \times 10^{-7}$$

$$v = 2.987 \times 10^4 \text{ m/s}$$

(iii) Centripetal acceleration

$$= \frac{v^2}{R} = \frac{(2.987 \times 10^4)^2}{1.5 \times 10^{11}}$$

$$= 5.948 \times 10^{-3} \text{ m/s}^2$$

Example 5.2 Calculate the mass of the earth from the following data.

$$g = 9.80 \text{ m/s}^2$$

$$G = 6.67 \times 10^{-11} \text{ M.K.S units}$$

$$R = 6.38 \times 10^6 \text{ m}$$

$$F = \frac{GMm}{R^2}$$

$$mg = \frac{GMm}{R^2} \text{ or } g = \frac{GM}{R^2}$$

$$M = \frac{gR^2}{G} = \frac{9.80 \times (6.38 \times 10^6)^2}{6.67 \times 10^{-11}}$$

$$M = 5.98 \times 10^{24} \text{ kg}$$

Example 5.3 The period of revolution of Phobos around Mars is 0.319 days and its mean distance from the Mars is $9.519 \times 10^8 \text{ m}$.

Calculate the mass of Mars (i) in kg and (ii) in units of earth's mass. Mass of the earth is 5.977×10^{24} kg.

Suppose mass of Mars = M

Mass of Phobos = m

$$M + m = \frac{4\pi^2}{G} \left(\frac{a^3}{T^2} \right)$$

Here, m can be neglected as compared to M

$$\therefore M = \frac{4\pi^2}{G} \left(\frac{a^3}{T^2} \right)$$

Here,

$$G = 6.66 \times 10^{-11} \text{ S.I. units}$$

$$a = 9.519 \times 10^6 \text{ m}$$

$$T = 0.319 \text{ days} = 0.319 \times 86400 \text{ s}$$

$$\therefore M = \frac{4\pi^2 (9.519 \times 10^6)^3}{6.66 \times 10^{-11} (0.319 \times 86400)^2}$$

$$M = 67.33 \times 10^{23} \text{ kg}$$

$$\text{Mass of the earth} = 5.977 \times 10^{24} \text{ kg}$$

$$\text{Mass of the Mars} = \left(\frac{6.733 \times 10^{23}}{5.977 \times 10^{24}} \right) \text{ mass of the earth}$$

$$= 0.1127 \text{ times the mass of the earth}$$

Example 5.4 The mean distance of the moon from the earth is 3.844×10^8 m and its period of revolution around the earth is 27.32 days. Calculate the mass of the earth and the moon, if the centre of mass of the earth-moon system is 4.75×10^6 m from the earth's centre.

$$\text{Mass of the earth} = M_1$$

$$\text{Mass of the moon} = M_2$$

$$a = 3.844 \times 10^8 \text{ m}$$

$$T = 27.32 \times 86400 \text{ s}$$

$$G = 6.66 \times 10^{-11} \text{ S.I. units}$$

$$M_1 + M_2 = \frac{4\pi^2 a^3}{GT^2}$$

$$M_1 + M_2 = \frac{4\pi^2 (3.844 \times 10^8)^3}{6.66 \times 10^{-11} \times (27.32 \times 86400)^2}$$

$$M_1 + M_2 = 6.045 \times 10^{24} \text{ kg}$$

also

$$a_1 = 4.75 \times 10^6 \text{ m}$$

$$a_1 + a_2 = a$$

$$a_1 + a_2 = 3.844 \times 10^8 \text{ m}$$

$$a_1 = 3.844 \times 10^8 - 4.75 \times 10^8 \text{ m}$$

$$a_2 = 379.65 \times 10^6 \text{ m}$$

$$M_1 a_1 = M_2 a_2$$

$$\frac{M_1}{M_2} = \frac{a_2}{a_1}$$

$$\frac{M_1 + M_2}{M_2} = \frac{a_1 + a_2}{a_1} = \frac{a}{a_1}$$

$$M_2 = \frac{(M_1 + M_2) a_1}{a}$$

$$M_2 = \frac{6.045 \times 10^{24} \times 4.75 \times 10^8}{3.844 \times 10^8}$$

$$M_2 = 7.469 \times 10^{24} \text{ kg} \quad (i)$$

$$\begin{aligned} M_1 &= 6.045 \times 10^{24} - 7.46 \times 10^{23} \\ &= 5.97 \times 10^{24} \text{ kg.} \end{aligned}$$

Example 5.4 (A). The maximum and minimum distances of a comet from the sun are $1.4 \times 10^{13} \text{ m}$ and $7 \times 10^{10} \text{ m}$ respectively. The velocity of the comet when it is nearest to the sun is $6 \times 10^4 \text{ m/s}$. What will be its velocity when it is farthest from the sun, assuming that in both positions, the comet is moving in circular orbits? [Delhi, 1980]

In this case

$$\frac{GM}{r_1^3} = \frac{v_1^2}{r_1} \quad \dots(1)$$

and

$$\frac{GM}{r_2^3} = \frac{v_2^2}{r_2} \quad \dots(2)$$

Dividing (1) by (2)

$$\frac{r_2^3}{r_1^3} = \left(\frac{v_1^2}{v_2^2} \right) \left[\frac{r_2}{r_1} \right]$$

$$v_2^2 = v_1^2 \left(\frac{r_1}{r_2} \right)$$

$$v_2 = v_1 \left[\frac{r_1}{r_2} \right]^{\frac{1}{2}}$$

Hence

$$v_1 = 6 \times 10^4 \text{ m/s}$$

$$r_1 = 7 \times 10^{10} \text{ m}$$

$$r_2 = 1.4 \times 10^{13} \text{ m}$$

$$v_2 = (6 \times 10^4) \left[\frac{7 \times 10^{10}}{1.4 \times 10^{13}} \right]^{\frac{1}{2}}$$

$$v_2 = 1.311 \times 10^4 \text{ m/s}$$

5.5 Determination of G by Cavendish Method

The first experimental method to determine the value of the gravitational constant G was devised by Cavendish in 1798. The arrangement consists of a strong metal bar PQ of length 183 cm and capable of turning about a vertical axis. The rod is suspended from the ceiling of a room. By operating the cords wound over the wheel W , the position of PQ can be altered from outside the room. C and D are two large lead spheres each of mass 150 kg and diameter 30 cm. These spheres are suspended to the ends of PQ by vertical metal rods. H is a torsion head which can be rotated from outside. The torsion head is directly below the middle of the rod PQ . A light rod RS of length about 180 cm is suspended with a silvered copper wire from the torsion head H . Two small spheres of mass 730 grams each are suspended from the ends of RS . Moreover, it is adjusted in such a way that the centres of all the four spheres, A , B , C , and D are in the same horizontal plane.

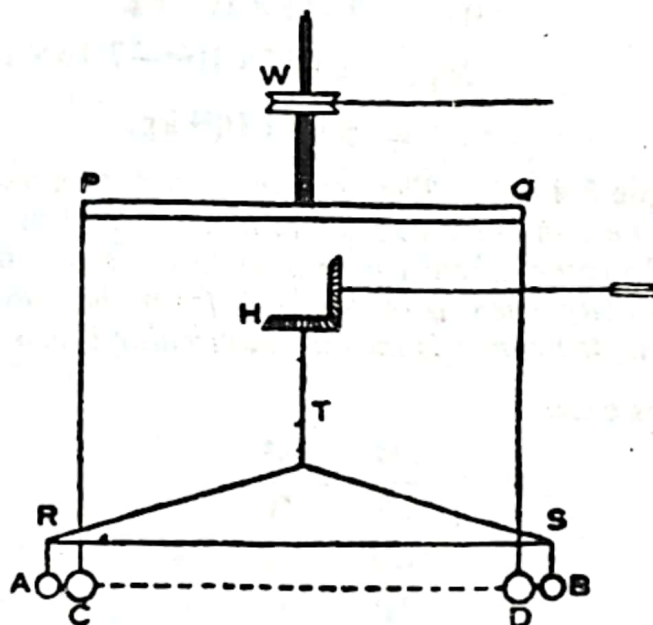


Fig 5.3

The ends of the torsion rod RS are provided with vernier scales which can slide on the main scales without touching the rod. The room is closed so as to prevent changes in temperature and circulation of air. The observations are made with the help of telescopes from outside the room through glass windows (Fig. 5.3).

Initially the rod PQ is rotated so that the line joining C and D is perpendicular to the line joining A and B (Fig. 5.4). In this position there is no torsion in the suspension wire T . The verniers are read.

Now, the rod PQ is rotated and brought to a position where C comes near A and D comes near B . The ball A is attracted towards C and B is attracted towards D . The force of attraction between a small ball and a bigger ball is given by

$$F = \frac{GMm}{d^2}.$$

in the equilibrium position, the readings of the verniers are noted. The deflecting couple is equal and opposite to the restoring couple due to the torsion in the wire.

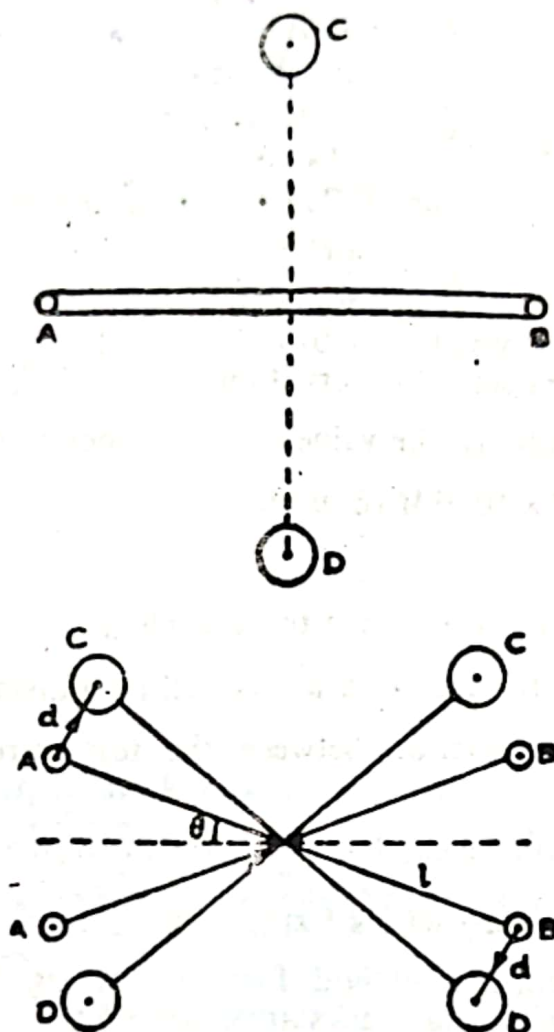


Fig. 5.4

$$\text{Deflecting couple} = \left(\frac{GMm}{d^2} \right) l \quad \dots(i)$$

Here l is the length of the torsion rod RS . If C is the restoring couple per unit twist and θ is the angle through which the torsion rod has been twisted, then

$$\text{Restoring couple} = C\theta \quad \dots(ii)$$

In the equilibrium position

$$\text{Deflecting couple} = \text{Restoring couple}$$

$$\therefore \frac{GMml}{d^2} = C\theta$$

$$G = \frac{Cd^2\theta}{Mml} \quad \dots(iii)$$

The experiment is repeated, by bringing C near B and D near A , keeping the distance d same. The value of θ is calculated in this case also. In equation (iii), the mean value of θ is used.

To find C , the torsion rod RS is set into torsional oscillations and the time period is noted. If I is the moment of inertia of the oscillating system,

$$t = 2\pi \sqrt{\frac{I}{C}}$$

$$\therefore C = \frac{4\pi^2 I}{t^2} \quad \dots(iv)$$

Substituting this value of C in equation, (iii) we get

$$G = \frac{4\pi^2 I d^2 \theta}{M m l t^2} \quad \dots(v)$$

The value of G was found to be 6.562×10^{-11} MKS units as a mean of 29 different sets of observations.

By recent methods, the value of G has been found to be

$$6.67 \times 10^{-11} \text{ MKS units.}$$

Corrections

Following corrections have to be applied :—

- (i) for the attraction of a bigger ball on a distant small ball
- (ii) for the attraction between the torsion rod and the two large spheres and
- (iii) for the forces exerted by the rods carrying the big balls.

Drawbacks in Cavendish's Experiment

In the Cavendish's method for determining the gravitational constant the following drawbacks are observed :

(1) As the force of attraction between any two balls is very small, to increase the deflecting couple, a long torsion rod has to be used. This requires a large chamber and convection currents cannot be avoided. Heavy traffic on a nearby road also causes vibrations in the torsion rod.

(2) As the suspension wire is thick, it requires a large-torque for unit angular displacement. In other words, for a given torque, the angle of deflection is very small.

(3) Due to its thickness, the torsion wire is not perfectly elastic. Hence, it does not completely return to the original position when the forces are withdrawn. Consequently the torque is not proportional to deflection.

(4) The angle of deflection is different from the observed value because of the attraction of the large spheres and their connecting rod.

(5) The measurement of the angle of deflection is not accurate.

5.9 Acceleration due to Gravity

Consider a body of mass m on the surface of the earth (Fig. 5.9). Mass of the earth = M and radius of the earth = R . The force of attraction exerted by the earth on the body according to Newton's law of gravitation,

$$F = \frac{GMm}{R^2}$$

According to Newton's second law of motion $F = ma$. Here $a = g$, where g is the acceleration due to gravity (Fig. 5.9).

$$\therefore F = ma$$

$$\text{or } mg = \frac{GMm}{R^2} \text{ or } g = \frac{GM}{R^2}$$

In CGS units, $g = 980 \text{ cm/s}^2$

In rationalised MKS units,

$$g = 9.80 \text{ m/s}^2$$

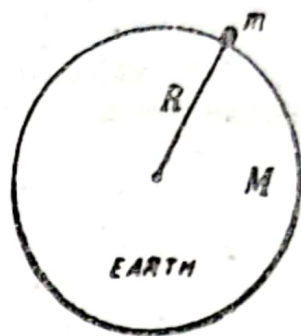


Fig 5.9

A body falling freely under the action of gravity towards the earth moves with a uniform acceleration equal to g .

Example 5.5 A body weighs 90 kg on the surface of the earth. How much will it weigh on the surface of the Mars whose mass is $1/9$ and radius $\frac{1}{2}$ that of the earth.

Mass of the earth = M , Mass of the body = m

Radius of the earth = R

From Newton's Law of Gravitation

$$F = \frac{G \times M \times m}{R^2}$$

$$F = 90 \text{ kg wt} = 90 \times 9.8 \text{ newtons}$$

$$90 \times 9.8 = \frac{G \times M \times m}{R^2} \quad \dots(i)$$

$$\text{Mass of the Mars} = \frac{M}{9}, \text{ Radius of the Mars} = \frac{R}{2}$$

Suppose the wt. of the body on the surface of Mars = x

$$\text{Then } x = \frac{G \times \frac{M}{9} \times m}{\left(\frac{R}{2}\right)^2} \quad \dots(ii)$$

Dividing (ii) by (i)

$$\begin{aligned} \frac{x}{90 \times 9.8} &= \frac{4}{9} \text{ or } x = \frac{90 \times 9.8 \times 4}{9} = 40 \times 9.8 \text{ newtons} \\ &= 40 \text{ kg wt} \end{aligned}$$

Thus, the body will weigh 40 kg wt on the surface of Mars.

Example 5.6 The mass of a planet is three times that of the earth and the diameter of the planet is also three times that of the earth. What is the acceleration due to gravity on the surface of the planet? Value of g on the earth's surface = 9.80 m/s^2

$$g = \frac{GM}{R^2} \quad \dots(i)$$

$$g' = \frac{GM'}{R'^2} \quad \dots(ii)$$

$$\therefore \frac{g'}{g} = \left(\frac{M'}{M}\right) \left(\frac{R^2}{R'^2}\right)$$

Here $\frac{M'}{M} = 3$

and $\frac{R}{R'} = \frac{1}{3}, g = 9.80 \text{ m/s}^2$

$$\therefore \frac{g'}{g} = 3 \times \frac{1}{9} = \frac{1}{3}$$

$$\therefore g' = \frac{g}{3} = \frac{9.8}{3} = 3.267 \text{ m/s}^2$$

Example 5.7 Calculate the radius of the moon's orbit assuming the orbit to be circular around the earth. Given $g = 9.8 \text{ m/s}^2$. Radius of the earth = $6.4 \times 10^6 \text{ m}$. Period of rotation of moon around the earth = 27.3 days. (Delhi, 1978)

Let the radius of the orbit be R and radius of the earth = r

$$\frac{g'}{g} = \frac{r^2}{R^2}$$

$$g' = \frac{gr^2}{R^2} \quad \dots(i)$$

Also $g' = \frac{v^2}{R}$

$$\therefore \frac{gr^2}{R^2} = \frac{v^2}{R} = \frac{\left(\frac{2\pi R}{T}\right)^2}{R} = \frac{4\pi^2 R}{T^2}$$

$$R^3 = \frac{gr^2 T^2}{4\pi^2}$$

$$R = \left[\frac{gr^2 T^2}{4\pi^2} \right]^{1/3}$$

$$R = \left[\frac{9.8 \times (6.4 \times 10^6)^2 (27.3 \times 86400)^2}{4 \times (3.142)^2} \right]^{1/3}$$

$$R = 3.838 \times 10^8 \text{ m}$$

5.10 Simple Pendulum

A simple pendulum consists of a light string supporting a small sphere and fixed firmly at its upper end. An ideal simple pendulum should consist of a heavy particle suspended by means of a weightless, inextensible, flexible string from a rigid support.

Let a pendulum be displaced from its mean position O and allowed to oscillate (Fig. 5.10). Suppose at any instant of time t , it is at A . The force acting upon the bob vertically downward $= Mg$. Resolve Mg into two rectangular components.

- (1) Force along the string $= Mg \cos \theta$
- (2) Force perpendicular to the string
 $= Mg \sin \theta$

Let the tension in the string be T . The component $Mg \cos \theta$ balances the tension T
 $Mg \cos \theta = T$

Thus the only force acting on the oscillating particle is $-Mg \sin \theta$.

$$\therefore F = -Mg \sin \theta$$

(-ve sign shows that the acceleration is directed towards the mean position)

According to Taylor's series of expansion

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

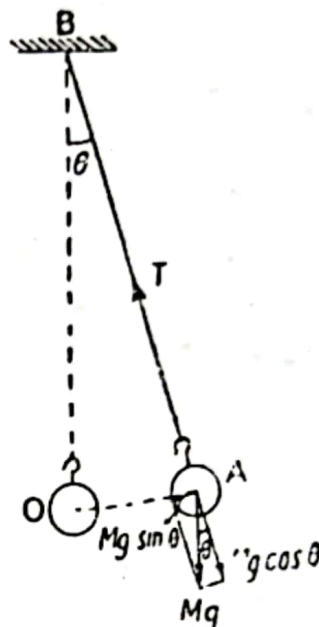


Fig. 5.10

For small angular displacements θ , $\sin \theta = \theta$

Tangential force $F = -Mg\theta$

The linear displacement $y = l\theta$

Acceleration $\frac{d^2y}{dt^2} = l \frac{d^2\theta}{dt^2}$

$$\therefore \text{Force} = Ml \frac{d^2\theta}{dt^2}$$

From Newton's second law

$$Ml \frac{d^2\theta}{dt^2} = -Mg\theta$$

$$-\frac{d^2\theta}{dt^2} + \left(\frac{g}{l}\right)\theta = 0 \quad \dots(1)$$

This equation is similar to the equation of simple harmonic motion

$$\frac{d^2y}{dt^2} + \omega^2 y = 0 \quad \dots(2)$$

From (1) and (2)

$$\omega^2 = \frac{g}{l}$$

$$\omega = \sqrt{\frac{g}{l}}$$

$$T = \frac{2\pi}{\omega}$$

$$\therefore T = 2\pi \sqrt{\frac{l}{g}} \quad \dots(3)$$

[Bob of large size. In the case of a simple pendulum, if the size of the bob is large, a correction has to be applied. In this case

$$T = 2\pi \sqrt{\frac{l + (\frac{2}{3} r^2)/l}{g}}$$

Here $l + \left(\frac{2}{3} \frac{r^2}{l}\right)$ represents the equivalent length of a simple pendulum].

TABLE 5.1
Value of g

Place	m/s ²	cm/s ²
Bombay	9.7863	978.63
Calcutta	9.7876	978.76
Delhi	9.7914	979.14
London	9.8117	981.17

The standard value of g now adopted is 980.665 cm/s² or 9.80665 m/s². This value is adopted as a conventional numerical value and does not correspond to any particular place.

Example 5.8 What is the maximum time period a simple pendulum can have on the surface of the earth?

From Fig. 5.11

$$F = mg \sin \theta$$

$$ma = mg \theta$$

$$a = g\theta$$

$$\text{But } \theta = \frac{Y}{R}$$

$$a = \frac{gY}{R}$$

$$\frac{\text{Displacement}}{\text{Acceleration}} = \frac{Y}{a} = \frac{R}{g}$$

$$t = 2\pi \sqrt{\frac{Y}{a}} = 2\pi \sqrt{\frac{R}{g}}$$

Radius of the earth

$$R = 6400 \text{ km}$$

$$g = 9.8 \text{ metres/s}^2$$

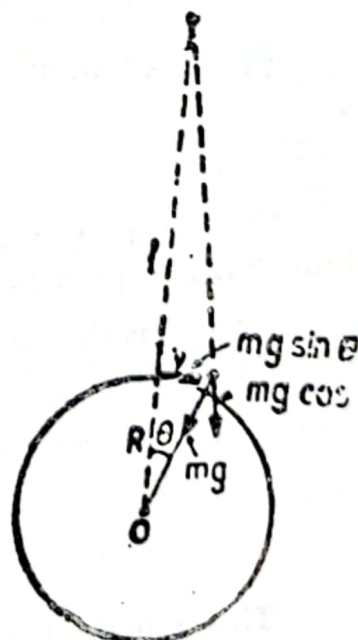


Fig. 5.11

$$t = 2\pi \sqrt{\frac{64 \times 10^5}{9.8}} = 5079 \text{ s}$$

$$= 1 \text{ hr } 24 \text{ min } 39 \text{ s.}$$

5.11 Compound Pendulum

A compound pendulum is a rigid mass capable of oscillating about a horizontal axis passing through any point of the mass. This point is called the point of suspension. In Fig. 5.12, G is the centre of gravity of the body and S is the point of suspension. At any instant of time, when the mass has been displaced, the force acting vertically downwards $= Mg$. At this position, the line SG makes an angle θ with the vertical and the restoring moment of this force about the point $S = Mg l \sin \theta$. This is the only moment which produces angular acceleration in the pendulum.

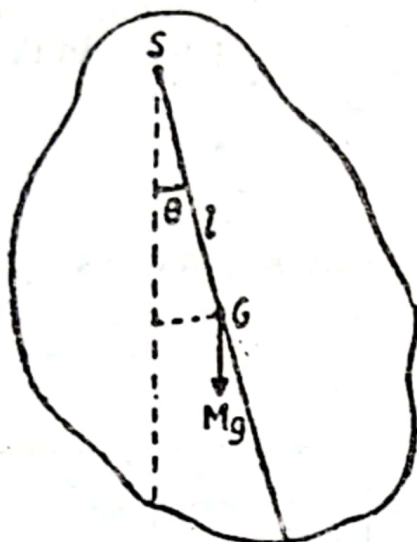


Fig. 5.12

Let the moment of inertia of the pendulum about an axis passing through S and perpendicular to its length be I . If the angular acceleration at this instant is

$$\frac{d^2\theta}{dt^2}, \text{ then}$$

$$I \frac{d^2\theta}{dt^2} = - Mg l \sin \theta$$

[—ve sign shows that the force is directed towards the mean position].

$$\therefore I \frac{d^2\theta}{dt^2} + Mg l \sin \theta = 0$$

For small angular displacements, $\sin \theta = \theta$

$$\therefore I \frac{d^2\theta}{dt^2} + Mg l \theta = 0$$

The moment of inertia of the pendulum about an axis passing through S and perpendicular to its plane $= MK^2 + MI^2$.

Here K is the radius of gyration about an axis passing through the centre of gravity of the pendulum.

$$\therefore M(K^2 + l^2) \frac{d^2\theta}{dt^2} + Mg l \theta = 0$$

$$\frac{d^2\theta}{dt^2} + \left(\frac{lg}{K^2 + l^2} \right) \theta = 0$$

This equation is similar to the equation of simple harmonic motion

$$\frac{d^2y}{dt^2} + \omega^2 y = 0 \quad \dots(2)$$

Here y refers to the angular displacement θ .

Comparing (1) and (2)

$$\omega^2 = \left(\frac{lg}{K^2 + l^2} \right)$$

Here ω is the angular frequency

$$\therefore \text{Time period } T = \frac{2\pi}{\omega}$$

$$T = 2\pi \sqrt{\frac{K^2 + l^2}{lg}} \quad \dots(3)$$

or

$$T = 2\pi \sqrt{\frac{(K^2/l) + l}{g}}$$

Here $(K^2/l + l)$ is called the equivalent length of the simple pendulum.

5.12 Bar Pendulum

A bar pendulum consists of a metal bar about 1 metre in length, 5 cm in breadth and 0.5 cm thick. It has a large number of holes as shown in Fig 5.13. There are two knife edges which fit into the holes. A bar pendulum is a particular case of a compound pendulum. The time period is determined by fixing the knife edge in each hole. The distance of each hole from the centre of gravity is measured. A graph is drawn between the distance from the C.G. along the X -axis and the corresponding time period along the Y -axis. The graph is as shown in Fig. 5.14.

From the graph the following results are obtained :

(1) On each side of the centre of gravity there are two positions for which the time periods are equal. It means there are four positions in the pendulum (two on each side of the C.G.) having the same time period. The points A, B, C and D in the graph have the same time period.

The time period of the bar pendulum,

$$t = 2\pi \sqrt{\frac{K^2 + l^2}{lg}}$$

Squaring

$$t^2 = 4\pi^2 \left(\frac{K^2 + l^2}{lg} \right)$$

or

$$4\pi^2 l^2 - g t^2 l + 4\pi^2 K^2 = 0 \quad \dots(1)$$

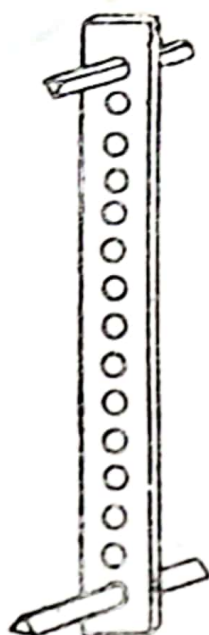


Fig. 5.13.

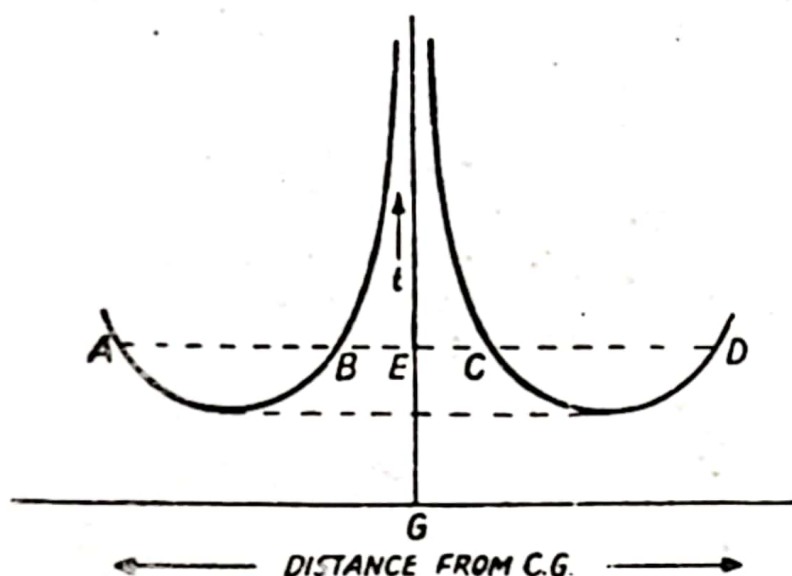


Fig. 5.14.

It is a quadratic equation in l

$$l = gt^2 \pm \sqrt{g^2 t^4 - 64\pi^4 K^2} \quad \dots(2)$$

From equation (2), the two values of l are

$$l_1 = gt^2 + \frac{\sqrt{g^2 t^4 - 64\pi^4 K^2}}{8\pi^2} \quad \dots(3)$$

and

$$l_2 = gt^2 - \frac{\sqrt{g^2 t^4 - 64\pi^4 K^2}}{8\pi^2} \quad \dots(4)$$

Thus two different points having the same value of t are obtained on each side of the bar pendulum.

(2) For a particular value of l , the time period is minimum, when

$$\sqrt{g^2 t^4 - 64\pi^4 K^2} = 0$$

$$l_1 = l_2 = \frac{gt^2}{8\pi^2} \quad \dots(5)$$

(3) It is seen from the graph that the time period increases enormously when the point of suspension approaches the centre of gravity.

From equation

$$t = 2\pi \sqrt{\frac{K^2 + l^2}{lg}}$$

When l is zero, t is infinite.

(4) From equations (3) and (4),

$$l_1 + l_2 = \frac{gt^2}{4\pi^2}$$

or

$$t = 2\pi \sqrt{\frac{l_1 + l_2}{g}} \quad \dots(6)$$

Here $(l_1 + l_2)$ is equivalent length of the simple pendulum.
In the graph

$$EA = ED = l_1$$

$$EB = EC = l_2$$

$$l_1 + l_2 = BA + EC = AC$$

Also $l_1 + l_2 = EB + ED = BD$

The value of g can be determined from equation (6)

Also $t = 2\pi \sqrt{\frac{K^2/l_1 + l_1}{g}} \quad \dots(7)$

Comparing equations (6) and (7)

$$\frac{K^2}{l_1} = l_2$$

or

$$K^2 = l_1 l_2$$

$$K = \sqrt{l_1 \times l_2}$$

also

$$K = \sqrt{EA \times EB} = \sqrt{ED \times EC}$$

Thus the value of radius of gyration K about the axis passing through the C.G. of a bar pendulum can be determined.

5.13 Points of Suspension and Oscillation are Interchangeable

In a bar pendulum, let the distance of A from the C.G. be l_1 and the distance of C be l_2 . The time period when A is the point of suspension is t_1 and the time period when C is point of suspension is t_2 .

The time periods are

$$t_1 = 2\pi \sqrt{\frac{K^2 + l_1^2}{l_1 g}} = 2\pi \sqrt{\frac{K^2/l_1 + l_1}{g}}$$

$$t_2 = 2\pi \sqrt{\frac{K^2 + l_2^2}{l_2 g}} = 2\pi \sqrt{\frac{K^2/l_2 + l_2}{g}}$$

But

$$\frac{K^2}{l_1} = l_2$$

$$\frac{K^2}{l_2} = l_1$$

$$\text{i.e.} \quad \frac{K^2}{l_1} + l_1 = \frac{K^2}{l_2} + l_2 = l_1 + l_2$$

$$\therefore \quad t_1 = t_2$$

Thus the points of suspension and oscillation are interchangeable. If A is the point of suspension, C is the point of oscillation. If C is the point of suspension, A is the point of oscillation.

5.14 Minimum Time Period

The time period of a compound pendulum,

$$t = 2\pi \sqrt{\frac{K^2 + l_1^2}{l_1 g}} \quad \dots(1)$$

Here l_1 is the distance of the point of suspension from the centre of gravity and K is the radius of gyration about the C. G. Suppose the distance of the point of oscillation from the C.G. is l_2 .

$$\therefore \quad \frac{K^2}{l_1} = l_2 \quad \text{or} \quad K^2 = l_1 l_2 \quad \dots(2)$$

For the time period to be minimum, the value of $\left(\frac{K^2}{l_1} + l_1\right)$ should be minimum.

Differentiating $\frac{K^2}{l_1} + l_1$ with respect to l_1

$$\frac{d}{dl_1} \left[\frac{K^2}{l_1} + l_1 \right] = -\frac{K^2}{l_1^2} + 1 \quad \dots(3)$$

This value should be zero for the time period to be minimum.

$$\therefore \quad -\frac{K^2}{l_1^2} + 1 = 0$$

or

$$K^2 = l_1^2$$

But

$$K^2 = l_1 l_2$$

$\therefore$

$$l_1 l_2 = l_1^2$$

or

$$l_2 = l_1$$

$\dots(4)$

It means that the time period will be minimum when the points of suspension and oscillation are equidistant from the centre of gravity.

[Note. Also t is maximum when l approaches zero, i.e. the point of suspension is at the centre of gravity, when

$$l = 0, t = \infty$$

Example 5.9. A circular disc is pivoted at its rim. Show that for small oscillations, its time period is given by

$$T = 2\pi \sqrt{\frac{3R}{2g}}$$

[Delhi, Engineering 1980]

Moment of inertia of a disc about an axis passing through its centre of gravity and perpendicular to its plane

$$= \frac{MR^2}{2}$$

$$\therefore MK^2 = \frac{MR^2}{2}$$

or $K^2 = \frac{R^2}{2}$

Time period of oscillation for a compound pendulum

$$T = 2\pi \sqrt{\frac{K^2 + l^2}{lg}}$$

Here

$$l = R$$

$$T = 2\pi \sqrt{\frac{\left(\frac{R^2}{2}\right) + R^2}{Rg}}$$

or $T = 2\pi \sqrt{\frac{3R}{2g}}$

Example 5.10. A heavy circular ring of radius R oscillates in a vertical plane about a horizontal axis at a distance x from the centre. Show that the time period is minimum when $x = R$

The radius of gyration for a circular ring about an axis passing through the C.G. and perpendicular to its plane.

$$K^2 = R^2$$

or

$$K = R$$

Here

$$l_1 = x$$

For the time period to be minimum

$$l_1 = l_2$$

or

$$K^2 = l_1 \times l_2 = l_1^2 = x^2$$

But

$$K^2 = R^2$$

$\therefore$

$$x^2 = R^2$$

or

$$x = R.$$

Taking

$$t_1 = t_2 = t \quad l_1 + l_2 = L$$

$$g = \frac{8\pi^2 L}{2t^2}$$

$$g = \frac{4\pi^2 L}{t^2} \quad \dots(9)$$

Here L is the distance between the two knife edges. Equation (9) is similar to the equation of a simple pendulum.

[Note. When the positions of K_1, K_2, M_1, M_2 and M_3 are finally adjusted, then the time periods about each knife edge must be equal. The positions of M_1, M_2 and M_3 must be the same while determining the time period about each knife edge.]

5.16. Value of g at the poles and at the Equator

The shape of the earth is slightly ellipsoidal. It is bulging at the equator and flattened at the poles. Its equatorial radius is more than the polar radius.

$$g = \frac{GM}{R^2}$$

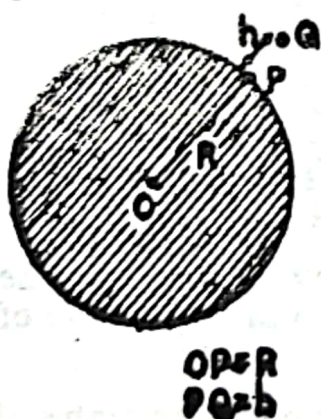
As

$$g \propto \frac{1}{R^2}$$

The value of g at the poles is greater as compared to its value at the equator.

5.17 Variation of g with Altitude

Let P be a point on the surface of the earth and Q another point at an altitude h (Fig. 5.15). R is the radius of the earth and M the mass of the earth. A body of mass m when placed at the point P experiences a force mg towards the centre of the earth.



$$\therefore mg = \frac{GMm}{R^2} \quad \dots(i)$$

When the body is at Q , let the acceleration due to gravity be g' .

$$mg' = \frac{GMm}{(R+h)^2} \quad \dots(ii)$$

Fig. 5.16

Dividing (ii) by (i)

$$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$$

$$g' = g \left[1 - \frac{2h}{R} \right] \quad \dots(iii)$$

Therefore, the acceleration due to gravity decreases with altitude.

5.18 Variation of g with Depth

Let g and g' be the accelerations due to gravity at P and Q respectively. At P the whole mass of the earth attracts the body and at Q it is attracted by the mass of the earth of radius $(R-h)$ (Fig. 5.17).

$$\therefore mg = \frac{GMm}{R^2} \quad \dots(i)$$

$$mg' = \frac{GM'm}{(R-h)^2} \quad \dots(ii)$$

Here $M = \frac{4}{3}\pi R^3 \rho$

$$M' = \frac{4}{3}\pi(R-h)^3 \rho$$

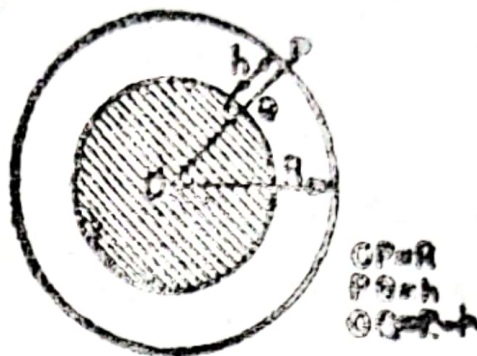


Fig. 5.17

where ρ is the mean density of the earth.

$\therefore$ Dividing (ii) by (i)

$$\frac{g'}{g} = \frac{M'}{M} \frac{R^2}{(R-h)^2}$$

$$\frac{g'}{g} = \frac{\frac{4}{3}\pi(R-h)^3 \rho}{\frac{4}{3}\pi R^3 \rho} \frac{R^2}{(R-h)^2}$$

$$g' = g \left[1 - \frac{h}{R} \right]$$

Therefore, the acceleration due to gravity decreases with depth.

5.19 Variation of g with Rotation of the Earth

Consider a body of mass m on the surface of the earth at a latitude ϕ .

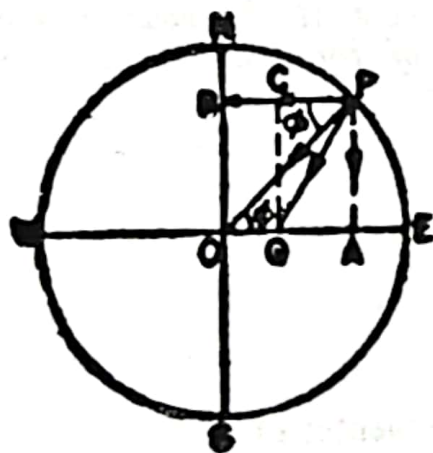


Fig. 5.18

Here $OP = R$

$\therefore BP = R \cos \phi$

Let ω be the angular velocity of the earth.

Force mg acts along PO . Resolve mg into two rectangular components (i) $mg \sin \phi$ along PA and (ii) $mg \cos \phi$ along PB (Fig. 5.18).

Due to the rotation of the earth, a body at P experiences an outward force

$$= \frac{mv^2}{r} = m\omega^2 r$$

$$r = BP = R \cos \phi$$

The net force along PC

$$= mg \cos \phi - mR\omega^2 \cos \phi$$

The resultant force experienced by P is along PQ which is the resultant of the force along PA and PC (represented vectorially).

$$\begin{aligned}\therefore F &= \sqrt{(mg \sin \phi)^2 + (mg \cos \phi - mR\omega^2 \cos \phi)^2} \\ &= mg \sqrt{1 + \frac{R^2 \omega^4 \cos^2 \phi}{g^2} - \frac{2R\omega^2 \cos^2 \phi}{g}}\end{aligned}$$

Here $\frac{R^2 \omega^4 \cos^2 \phi}{g^2}$ is negligible

$$\therefore F = mg \sqrt{1 - \frac{2R\omega^2 \cos^2 \phi}{g}}$$

$$\therefore mg' = mg \left(1 - \frac{R\omega^2 \cos^2 \phi}{g} \right)$$

$$\therefore g' = g \left(1 - \frac{R\omega^2 \cos^2 \phi}{g} \right) \quad \dots(i)$$

Special cases

(1) At the equator, $\phi = 0$, $\cos \phi = 1$

$$g' = g \left(1 - \frac{R\omega^2}{g} \right)$$

But

$$\frac{R\omega^2}{g} = \frac{1}{289}$$

$$g' = g \left(1 - \frac{1}{289} \right) = \frac{288}{289} g$$

(2) At the poles $\phi = 90$, $\cos \phi = 0$

$$\therefore g' = g$$

Therefore the acceleration due to gravity at the poles is greater than that at the equator.

Example 5.15 Show that the objects lying at the equator will fly off the surface of the earth, if the speed of rotation of the earth increases seventeen times its present value.

At the equator

$$g' = g \left(1 - \frac{R\omega^2}{g} \right)$$

Here

$$\frac{R\omega^2}{g} = \frac{1}{289}$$

When the speed of rotation increases seventeen times

$$g' = g \left(1 - \frac{R(17\omega)^2}{g} \right)$$

$$g' = g \left[1 - 289 \frac{R\omega^2}{g} \right]$$

$$g' = g \left[1 - \frac{289}{289} \right] = 0$$

$$g' = 0$$

Therefore, a body lying at the equator will fly off because it experiences no force toward the centre of the earth due to gravity.

5.20 Difference Between Mass and Weight

Mass is the amount of matter contained in a body. This is fixed for a given body. Its value is the same at all places. Weight is the force with which a body is attracted towards the centre of the earth. It is different at different places. It depends on the value of g . A body has the same mass at the poles and at the equator whereas its weight will be more at the poles than at the equator.

With the help of a beam balance, the mass of a body is determined. With the help of a spring balance the weight of the body is determined.

The units of mass are g and kg.

The units of weight are dyne, newton, g wt and kg wt.

5.21 Inertial Mass and Gravitational Mass

Inertial Mass: According to Newton's second law of motion, when a force is applied on a body, the body moves with an acceleration

$$F = ma \quad \text{or} \quad m = \frac{F}{a}$$

The mass is called the inertial mass of the body. If the force is increased, the acceleration also increases and $\frac{F}{a} = \text{constant}$ for a given body. If the same force is applied on two different bodies, and the accelerations produced are equal, then the inertial masses of the two bodies are equal. If the same force is applied on two different bodies, the inertial mass of that body is more in which the acceleration produced is less and *vice versa*.

Gravitational Mass: According to the law of gravitation, the gravitational force of attraction of a body towards the centre of the earth is equal to the weight of the body.

Let the weight of the body be W

$$W = mg$$

$$m = \frac{W}{g}$$

This mass is called the gravitational mass of the body. This is determined with the help of a beam balance. It will be the same even on the surface of the moon. If the value of g is less at the moon,

the weight of the body will also be less but the gravitational mass is the same.

Note. There is no relation between the inertial and the gravitational mass of a body, but for all practical purposes, both are taken to be identical and their units are the same.

5.22 Gravitational Field

The space around a body (of mass M) within which the gravitational force of attraction is perceived is called gravitational field. The gravitational field or intensity at a point is the force experienced by a unit mass at that point. If the gravitational field at a point is E , the force acting on a mass m is F ,

$$F = mE$$

$$E = \frac{F}{m} \quad \dots(i)$$

Also the gravitational field is defined as the negative gradient of gravitational potential

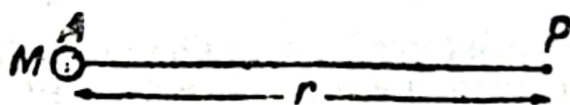
$$\therefore E = -\frac{dV}{dx}$$

5.23 Gravitational Potential

Consider a particle A of mass m . P is a point at a distance r from A . The gravitational intensity at P ,

$$E = \frac{F}{m} = \frac{GM}{r^2} \quad \dots(i)$$

Gravitational potential at a point is defined as the amount of work done in moving a unit mass from the point to infinity against the gravitational force of attraction (Fig. 5.19).



(Fig. 5.19)

$$E = -\frac{dV}{dr} \quad \dots(ii)$$

or

$$dV = -E \times dr$$

$$dV = \left(\frac{-GM}{r^2} \right) dr$$

Integrating between the limits infinity and r ,

$$\int dV = \int_r^\infty \left(\frac{-GM}{r^2} \right) dr = \left[\frac{GM}{r} \right]_r^\infty = \left[\frac{GM}{\infty} - \frac{GM}{r} \right]$$

$$V = -\frac{GM}{r} \quad \dots(iii)$$

The gravitational potential at a point due to a point mass

$$= \left(-\frac{GM}{r} \right)$$

It is maximum and zero at infinity. At all other points it is less than zero i.e. a negative value. Gravitational potential at a point is a scalar quantity.

5.24 Gravitational Potential Energy

The force of gravitation is always a force of attraction. Consider a body of mass M at the point A . Take a point P at a distance r from A (Fig. 5.19). If a body of mass m is placed at P , it experiences a force of attraction towards A . If this body is to be moved from A , work has to be done. This work done is stored in the body as its gravitational potential energy. The gravitational potential of the body of mass m at the point P will be equal to the product of the mass and the gravitational potential at the point P .

$$\therefore \text{Potential energy} = m \left(-\frac{GM}{r} \right)$$

Thus gravitational potential energy of a body depends upon

- (i) the mass of the body at A
- (ii) the mass of the body at P
- (iii) the distance between two masses.

At infinity, the gravitational potential energy of a body is maximum and zero. At all other points, it is negative.

5.25 Gravitational Potential and Field at a Point due to a Spherical Shell

- (i) At a point outside the shell

Consider a uniform spherical shell of mass M and radius a (Fig. 5.20). Let ρ be the density per unit area of the shell. The planes AC and BD cut the shell vertically and the element between the two planes is a ring of radius $AB = a \sin \theta$.

Surface area of the element

$$= 2\pi (a \sin \theta) a d\theta = 2\pi a^2 \sin \theta d\theta$$

Mass of the element, $m = (2\pi a^2 \sin \theta d\theta) \rho$

Also

$$AP = x$$

Potential at P due to the element,

$$dV = -\frac{Gm}{x}$$

$$dV = -\frac{G \rho a^2 \sin \theta d\theta}{x} \quad \dots(i)$$

In the $\triangle OAP$

$$x^2 = a^2 + R^2 - 2aR \cos \theta$$

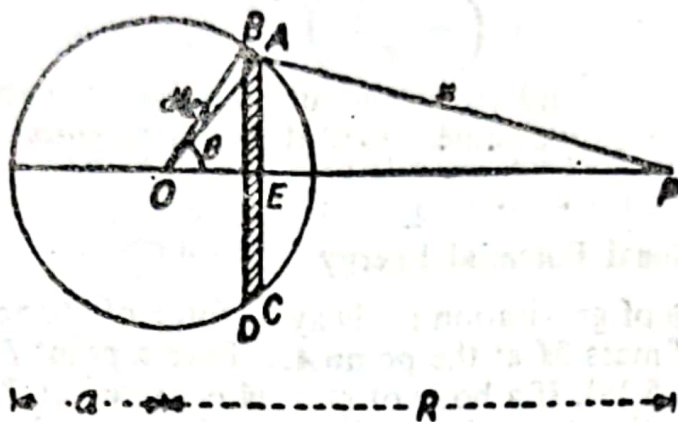


Fig. 5.20

Differentiating,

$$2xdx = 2aR \sin \theta d\theta$$

$$x = \frac{aR \sin \theta d\theta}{dx}$$

Substituting this value of x in equation (i)

$$dV = - \frac{G 2\pi a^2 \rho \sin \theta d\theta dx}{aR \sin \theta d\theta}$$

$$dV = - \left(\frac{2\pi Ga\rho}{R} \right) dx \quad \dots(ii)$$

Integrating for the whole shell

$$V = - \int_{R-a}^{R+a} \left(\frac{2\pi Ga\rho}{R} \right) dx$$

$$V = - \frac{4\pi a^2 \rho G}{R}$$

But

$$M = 4\pi a^2 \rho$$

$$\therefore V = - \frac{GM}{R} \quad \dots(iii)$$

Thus for a point outside the shell, the shell behaves as if the whole of its mass is concentrated at the centre of the shell.

Further

$$V \propto \frac{1}{R} \quad \dots(iv)$$

Gravitational Field

$$E = - \frac{dV}{dR}$$

$$E = - \frac{d}{dR} \left[- \frac{GM}{R} \right]$$

$$\vec{E} = - \frac{GM \hat{R}}{R^2}$$

The negative sign indicates that the field is towards the centre of the shell.

(ii) At a point inside a shell

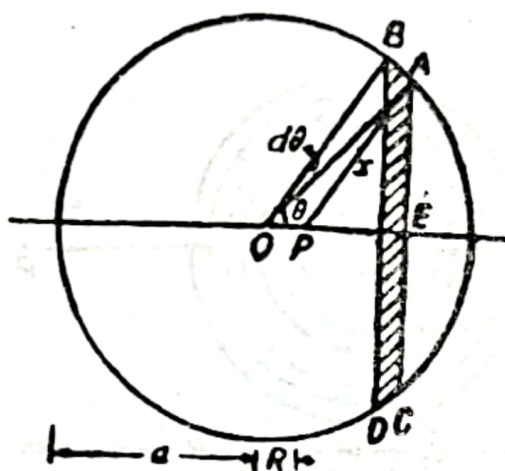


Fig. 5.21

Potential at a point inside the shell, due to the element (Fig. 5.21),

$$dV = - \left(\frac{2\pi G a \rho}{R} \right) dx$$

For the whole shell, integrate between the limits $(a-R)$ and $(a+R)$

$$\therefore V = \int_{a-R}^{a+R} - \left(\frac{2\pi G a \rho}{R} \right) dx$$

$$V = - G 4 \pi a \rho$$

But

$$M = 4 \pi a^2 \rho$$

$$V = - \left(\frac{GM}{a} \right) \quad \dots(v)$$

Expression (v) shows that V is independent of R . It is the same at all points inside the shell. This potential is equal to the potential on the surface of the shell and is constant.

Gravitational Field

$$E = - \frac{dV}{dR}$$

$$E = - \frac{d}{dR} \left[- \frac{GM}{a} \right]$$

As $\left(-\frac{GM}{a} \right)$ is a constant,

$$E = 0$$

Therefore the gravitational field at any point inside a spherical shell is zero.

5.26 Gravitational Potential and Field due to a Uniform Solid Sphere

(i) At a point outside the sphere

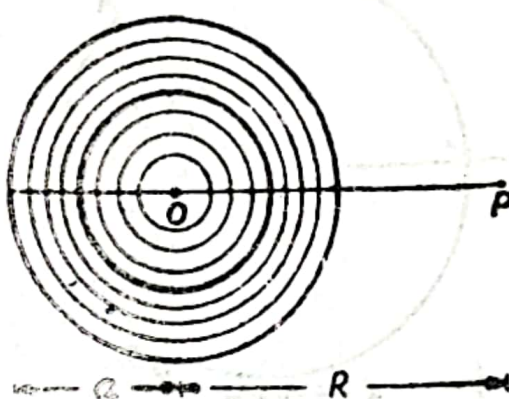


Fig. 5.22

The sphere can be considered to be made up very large number of spherical shells whose masses are $m_1, m_2, m_3, \dots$ etc. (Fig. 5.22) The potential at the point P due to all the shells

$$V = - \frac{G}{R} [m_1 + m_2 + m_3 + \dots]$$

$$V = - \frac{GM}{R} \quad \dots(i)$$

Gravitation field

$$E = - \frac{dV}{dR}$$

$$E = - \frac{d}{dR} \left[\frac{GM}{R} \right]$$

$$E = - \frac{GM}{R^2} \quad \dots(ii)$$

Negative sign indicates that the field is towards the centre of the sphere.

(ii) At a point inside the sphere

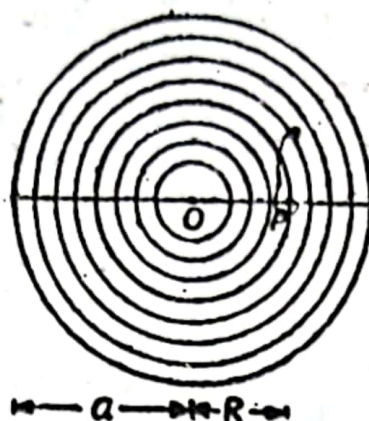


Fig. 5.23

The solid sphere can be considered to be made up of a large number of thin spherical shells (Fig. 5.23). Radius of the sphere $= a$ and its density $= \rho$

The point P lies outside the shells of radii less than R and P lies inside the shells of radii varying from R to a .

The gravitational potential at P is equal to the sum of the potential due to these two sets of shells.

Potential at P due to shells of radii less than R ,

$$V_1 = - \left[\frac{G \frac{4}{3} \pi R^3 \rho}{R} \right]$$

$$V_1 = - \frac{4}{3} \pi G R^2 \rho \quad \dots(i)$$

Potential at P due to the shells of radii varying from R to $a = V_2$

To calculate V_2 consider a spherical shell of radius x and radial thickness dx .

$$\text{Volume of the shell} = 4\pi x^2 dx$$

$$\text{Mass of the shell} = (4\pi x^2 dx) \rho = 4\pi \rho x^2 dx$$

$$\text{Potential at } P \text{ due to this shell} = - \frac{G 4\pi \rho x^2 dx}{x}$$

$$dV = -4\pi G \rho x dx$$

$$V_2 = \int_R^a dV = -4\pi G \rho \int_R^a x dx$$

$$V_2 = -2\pi G \rho (a^2 - R^2) \quad \dots(ii)$$

Total potential at P ,

$$V = V_1 + V_2$$

$$\begin{aligned}
 V &= -\frac{4}{3}\pi GR^2\rho - 2\pi G\rho(a^2 - R^2) \\
 V &= -G\frac{4}{3}\pi a^3\rho \left[\frac{R^2}{a^3} + \frac{3}{2a} - \frac{3R^2}{2a^3} \right] \\
 V &= -GM \left[\frac{3a^2 - R^2}{2a^3} \right] \quad \dots(iii)
 \end{aligned}$$

Gravitational Field

$$\begin{aligned}
 E &= -\frac{dV}{dR} \\
 E &= -\frac{d}{dR} \left[-\frac{GM(3a^2 - R^2)}{2a^3} \right] \\
 E &= -\frac{GM\hat{R}}{a^3} \quad \dots(iv)
 \end{aligned}$$

Negative sign indicates that the field is towards the centre of the sphere.

5.27 Gravitational Potential and Field due to a Hollow Sphere

(i) At a point outside the sphere.

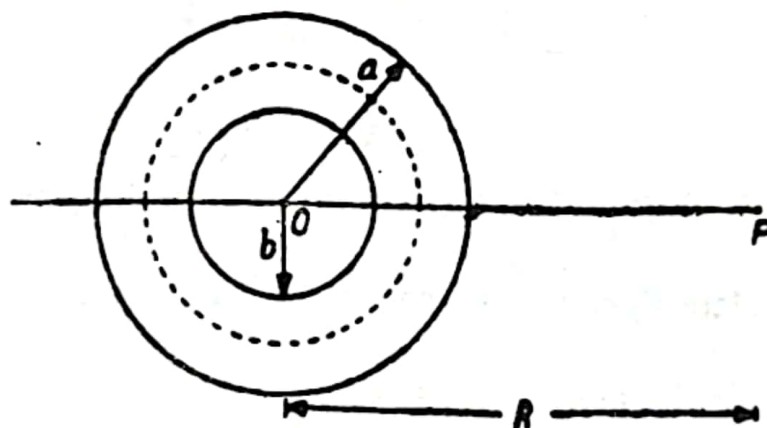


Fig. 5.24

Consider the sphere to be made up of a large number of thin spherical shells of radii varying from b to a (Fig. 5.24). At the point P outside the shell.

$$\begin{aligned}
 V &= -\frac{G}{R}(m_1 + m_2 + \dots) \\
 V &= -\frac{GM}{R}
 \end{aligned}$$

Gravitational Field

$$E = -\frac{dV}{dR} = -\frac{d}{dR} \left(-\frac{GM}{R} \right)$$

$$E = - \frac{GMR}{R^2} \quad \dots(ii)$$

(ii) At a point inside the hollow space.

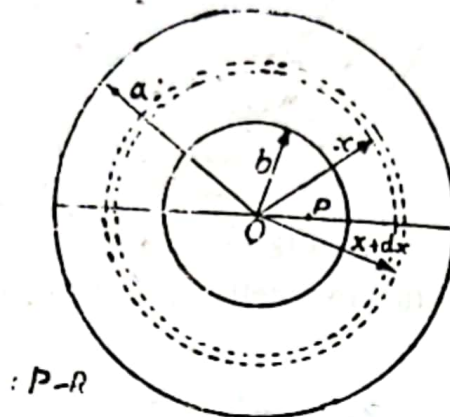


Fig. 5.25

Consider a spherical shell of radius x and radial thickness dx .
(Fig 5.25)

$$\text{Volume of the shell} = 4\pi x^2 dx$$

$$\text{Mass of the shell} = 4\pi \rho x^2 dx$$

Potential at P due to this shell,

$$dV = - \frac{G4\pi \rho x^2 dx}{x} = -4\pi \rho x dx$$

Potential at P due to the whole hollow sphere

$$V = -4\pi G\rho \int_b^a x dx$$

$$V = -2\pi \rho G (a^2 - b^2)$$

But

$$M = \frac{4}{3}\pi (a^3 - b^3) \rho$$

$$\therefore V = -\frac{3}{2} GM \left(\frac{a^2 - b^2}{a^3 - b^3} \right) = -\frac{3GM(a+b)}{2(a^2 + ab + b^2)}$$

Gravitational Field

As a and b are constant, V is constant

$$E = - \frac{dV}{dR} = 0$$

(iii) At a point in between the two radii (inside the material),

Here

$$b < R < a$$

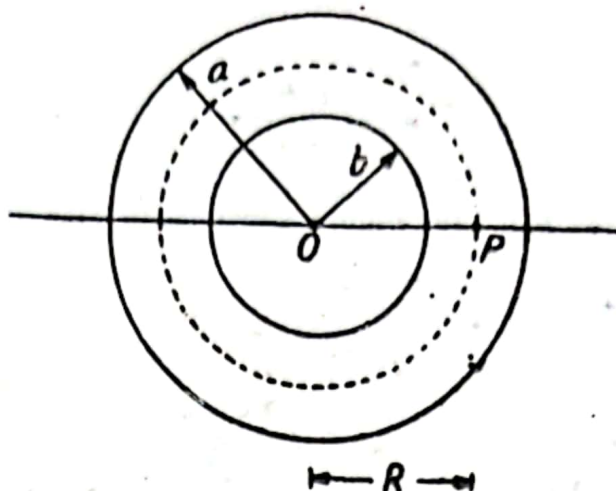


Fig. 5.26

Potential at P due to shells of radii varying from b to R ,
(Fig. 5.26)

$$V_1 = -\frac{Gm_1}{R},$$

But $m_1 = \frac{4}{3}\pi(R^3 - b^3)\rho$

Potential at P due to shells of radii varying from R to a ,

$$V_2 = -2\pi G\rho(a^2 - R^2)$$

Therefore, total potential at P ,

$$V = V_1 + V_2$$

$$V = -\frac{\frac{4}{3}\pi G\rho(R^3 - b^3)}{R} - 2\pi G\rho(a^2 - R^2)$$

$$V = -\frac{4}{3}\pi\rho(a^3 - b^3)\frac{G}{2R}\left[\frac{2(R^3 - b^3) + 3(a^2 - R^2)R}{(a^3 - b^3)}\right]$$

$$V = -\frac{GM}{2R}\left[\frac{3a^3R - R^3 - 2b^3}{(a^3 - b^3)}\right] \quad \dots(i)$$

Field

$$E = -\frac{dV}{dR}$$

$$E = -\frac{d}{dR}\left[-\frac{GM}{2(a^3 - b^3)}\left(3a^3 - R^3 - \frac{2b^3}{R}\right)\right]$$

$$E = -\frac{GM}{2(a^3 - b^3)}\left[2R - \frac{2b^3}{R^2}\right]$$

$$E = -\frac{GM}{R^2}\left(\frac{R^3 - b^3}{a^3 - b^3}\right)\hat{R} \quad \dots(ii)$$

Negative sign indicates that the field is towards the centre of the sphere.

5.28 Gravitational Potential and Field due to a Thin Circular Plate

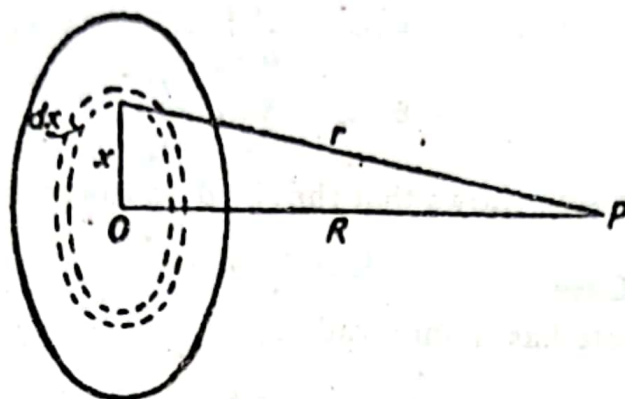


Fig. 5.27

Consider a circular plate of radius a and surface density ρ ,
(Fig 5.27)

The mass of the plate

$$M = \pi a^2 \rho$$

The point P lies on the axis passing through the centre and perpendicular to the plane of the disc. The distance

$$OP = R.$$

Consider an element of radius x and radial thickness dx at a distance r from P .

$$\text{Mass of the element} = 2\pi \rho x dx$$

Potential at P due to the element,

$$dV = - \frac{G(2\pi \rho x dx)}{r}$$

$$dV = - \frac{2\pi G \rho x dx}{(x^2 + R^2)^{1/2}}$$

Potential at P due to the whole plate,

$$V = -2\pi G \rho \int_0^a \frac{x dx}{(x^2 + R^2)^{1/2}}$$

$$V = -2\pi G \rho \left[(x^2 + R^2)^{1/2} \right]_0^a$$

$$V = -2\pi G \rho \left[(a^2 + R^2)^{1/2} - R \right] \quad \dots(i)$$

Let

$$(a^2 + R^2) = y^2$$

$$V = -2\pi G \rho [y - R] \quad \dots(ii)$$

Gravitational Field

$$E = - \frac{dV}{dR}$$

$$E = - \frac{d}{dR} \left[-2\pi G \rho (\sqrt{a^2 + R^2} - R) \right]$$

$$E = - 2\pi G \rho \left[1 - \frac{R}{(a^2 + R^2)^{1/2}} \right] \dots(iii)$$

Negative sign shows that the field is towards the centre of the plate.

Special Case

If the plate has infinite radius,

$$a = \infty \text{ and } \frac{R}{(a^2 + R^2)^{1/2}} = 0$$

$$E = -2\pi G \rho \dots(iii)$$

5.29 Satellites

In the solar system, different planets revolve round the sun. The radii of the orbits and their time periods of revolution are different for different planets. In these cases, the force of gravitation between the sun and the planet provides the necessary centripetal force. Similarly the moon revolves around the earth and the force of gravitation between the earth and the moon provides the necessary centripetal force for the moon to be in the orbit. Here moon is the satellite of the earth.

From 1957, many artificial satellites around the earth have been launched. These satellites are put into orbits with the help of multi-stage rockets.

Orbital Velocity. Suppose a satellite of mass m is moving in an orbit of radius r around the earth. If v is the velocity of the satellite in its orbit, the necessary centripetal force $\left(\frac{mv^2}{r}\right)$ is provided by the gravitational force of attraction between the earth and the satellite.

$$\text{Gravitational force} = \frac{GMm}{r^2}$$

Here, M is the mass of the earth

$$\therefore \frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$\text{or } v = \sqrt{\frac{GM}{r}} \dots(i)$$

If g is the acceleration due to gravity at the position of the satellite,

$$mg = \frac{GMm}{r^2}$$

or $\frac{GM}{r} = gr \quad \dots(ii)$

From equations (i) and (ii)

$$\therefore v = \sqrt{gr}$$

For a satellite, revolving very near the earth's surface,

$$v = \sqrt{gr}$$

$$\therefore v = \sqrt{gR} \quad \dots(iii)$$

Taking the value of

$$g = 9.80 \text{ m/s}^2$$

and

$$R = 64 \times 10^5 \text{ m}$$

$$v = \sqrt{9.80 \times 64 \times 10^5}$$

$$= 7,920 \text{ m/s}$$

$$= 7.920 \text{ km/s.}$$

Time Period

From equation (i)

$$v = \sqrt{\frac{GM}{r}}$$

But

$$v = r\omega$$

$$\therefore r\omega = \sqrt{\frac{GM}{r}}$$

$$\omega = \sqrt{\frac{GM}{r^3}} \quad \dots(iv)$$

The time period of revolution of the satellite is given by

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{r^3}{GM}}$$

But

$$r = R + h$$

$$T = 2\pi \sqrt{\frac{(R+h)^3}{GM}}$$

From equation (ii) $GM = gr^2 = g(R+h)^2$

$$\therefore T = 2\pi \sqrt{\frac{(R+h)^3}{g(R+h)^2}}$$

$$T = 2\pi \sqrt{\frac{R+h}{g}} \quad \dots(v)$$

When $h < R$,

$$T = 2\pi \sqrt{\frac{R}{g}} \quad \dots(vi)$$

5.30 Stationary Satellite

Consider a satellite in a circular orbit round the earth. It is coplanar with the equator of the earth. Its period of rotation is the same as that of the earth. This satellite appears to be stationary to the observer on the earth. Such satellites are used widely for television and radio broadcasting.

The angular velocity of this satellite is given by

$$\omega = \frac{2\pi}{T}, \text{ Here } T = 24 \text{ hours} = 86400 \text{ s}$$

$$\omega = \frac{2\pi}{86400} = 7.3 \times 10^{-5} \text{ radians/s}$$

The distance of such a satellite from the centre of the earth is given by

$$\omega = \sqrt{\frac{GM}{r^3}}$$

$$r = \left(\frac{GM}{\omega^2}\right)^{1/3} = \left[\frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{(7.3 \times 10^{-5})^2}\right]^{1/3}$$

$$r = 4.2 \times 10^7 \text{ m}$$

This distance is about 6.6 times the radius of the earth. The distance of such a satellite from the surface of the earth is

$$h = r - R = 4.2 \times 10^7 - 6.4 \times 10^6$$

$$h = 35.6 \times 10^6 \text{ m}$$

$$h = 3.56 \times 10^7 \text{ m}$$

Example 5.16 A satellite goes round the earth in 90 minutes in a circular orbit. Calculate the height of the satellite above the earth taking the earth to be a sphere of radius 6370 km. The value of g at the orbit of satellite = 9.80 m/s^2 .

$$\frac{mv^2}{r} = mg$$

$$\frac{v^2}{r} = g$$

But

$$v = r\omega$$

$$\omega = \frac{2\pi}{T}$$

$\therefore$

$$g = \frac{r^2\omega^2}{r} = \frac{4\pi^2 r}{T^2}$$

Here

$$g = 9.8 \text{ m/s}^2, \quad T = 90 \times 60 = 5400 \text{ s}$$

$\therefore$

$$r = \frac{gT^2}{4\pi^2} = \frac{9.8 \times (5400)^2 \times 7 \times 7}{4 \times 22 \times 22}$$

$$r = 7235 \times 10^3 \text{ m}$$

$$r = 7235 \text{ km}$$

Since the radius of the earth = 6370 km, the height of the satellite above the earth's surface

$$= 7235 - 6370 = 865 \text{ km}$$

Example 5.17 With what velocity should a body be projected horizontally at a height of 30 km from the ground so that it becomes the satellite of the earth? Neglect frictional losses. Also calculate the time period of revolution of the satellite. Radius of the earth = 6370 km.

Let the radius of the earth be R and the velocity with which the body is projected be v .

Here $R = 6370 \text{ km}$ and $h = 30 \text{ km}$

$$(i) \frac{mv^2}{(R+h)} = mg' \text{ where } g' = g \left(1 - \frac{2h}{R} \right)$$

$$v^2 = g' (R+h) = g (R+h) \left(1 - \frac{2h}{R} \right)$$

$$v = \sqrt{gR \left(1 - \frac{h}{R} \right)}$$

$$= \sqrt{9.8 \times 6370 \times 10^3 \left(1 - \frac{30}{6370} \right)}$$

$$= 7918 \text{ m/s} = 7.918 \text{ km/s}$$

$$(ii) \quad t = \frac{2\pi(R+h)}{v} = \frac{2\pi(R+h)}{\sqrt{gR \left(1 - \frac{h}{R} \right)}}$$

$$t = 2\pi \sqrt{\frac{R^2 \left(1 + \frac{h}{R} \right)}{gR \left(1 - \frac{h}{R} \right)}} = 2\pi \sqrt{\frac{R \left(1 + \frac{3h}{R} \right)}{g}}$$

$$= 2\pi \sqrt{\frac{6370 \times 10^3 \left(1 + \frac{3 \times 30 \times 10^3}{6370 \times 10^3} \right)}{9.8}}$$

$$= 5102 \text{ s} = 1 \text{ hr } 25 \text{ min } 2 \text{ s}$$

5.31 Escape Velocity

Escape velocity is defined as the velocity with which a body has to be projected vertically upwards from the earth's surface so that it escapes the earth's gravitational field altogether.

If v is the escape velocity, then the initial kinetic energy of projection $\frac{1}{2}mv^2$ must be equal to the work done in moving the body

from the surface of the earth to infinity. In such a case, the body does not return to the earth's surface.

$$\text{Mass of the earth} = M$$

$$\text{Radius of the earth} = R$$

$$\text{Constant of Gravitation} = G$$

$$\text{Mass of the body} = m$$

Suppose a body is at a distance x from the earth's centre. The force of gravity acting on the body

$$= \frac{GMm}{x^2}$$

If the body is to be moved through a small distance dx away from the earth's surface

$$\text{Work done, } W = \left(\frac{GMm}{x^2} \right) dx \quad \dots(i)$$

Work done in moving the body from the surface of the earth to infinity

$$\int dW = \int_R^\infty \frac{GMm}{x^2} \cdot dx$$

$$W = \left[\frac{-GMm}{x} \right]_R^\infty$$

$$W = \left[\frac{GMm}{R} \right] \quad \dots(ii)$$

$$K.E. = \frac{1}{2}mv^2 \quad \dots(iii)$$

$$\therefore \frac{1}{2}mv^2 = \frac{GMm}{R}$$

$$v = \sqrt{\frac{2GM}{R}} \quad \dots(iv)$$

Also at the surface of the earth

$$mg = \frac{GMm}{R^2}$$

$$\text{or } GM = gR^2 \quad \dots(v)$$

Substituting this value of GM in equation (iv)

$$v = \sqrt{2gR} \quad \dots(vi)$$

The value of g at the earth's surface = 9.80 m/s^2 and

$$R = 64 \times 10^5 \text{ m}$$

$$v = \sqrt{2 \times 9.80 \times 64 \times 10^5}$$

$$v = 11.2 \times 10^3 \text{ m/s}$$

$$v = 11.2 \text{ km/s}$$

If a body is projected vertically upwards with a velocity of 11.2 km/s or more, it will not return to the earth's surface.

Example 5.18. With what velocity should a body be projected vertically upwards from the surface of the earth so that it may just attain a height of $R/2$, where R is the radius of the earth.

($R = 6400$ km)

[Delhi, Engineering 1980]

Work done in moving the body from the surface of the earth to a height $R/2$ above the earth's surface

$$W = \int_R^{3R/2} \left(\frac{GMm}{x^2} \right) dx$$

$$W = \left[-\frac{GMm}{x} \right]_R^{3R/2}$$

$$W = GMm \left[\frac{1}{3R/2} - \frac{1}{R} \right]$$

$$W = \frac{GMm}{3R}$$

$$KE = \frac{1}{2} mv^2$$

$$\therefore \frac{1}{2} mv^2 = \frac{GMm}{3R} \quad \dots(i)$$

At the surface of the earth

$$mg = \frac{GMm}{R^2} \quad \dots(ii)$$

Dividing (i) by (ii)

$$\frac{v^2}{2g} = \frac{R}{3}$$

$$v = \sqrt{\frac{2gR}{3}}$$

Here $g = 9.80 \text{ m/s}^2$ and $R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$

$$\therefore v = \sqrt{\frac{2 \times 9.80 \times 6.4 \times 10^6}{3}}$$

$$v = 6.467 \times 10^3 \text{ m/s} = 6.467 \text{ km/s}$$

Examples 5.19. Calculate the mass of the sun, given that the distance between the sun and the earth is $1.49 \times 10^{11} \text{ m}$ and $G = 6.66 \times 10^{-11} \text{ SI units}$, 1 year = 365 days, [Delhi, 1976]

$$\frac{m^2}{r} = \frac{GMm}{r^2}$$

$$v^2 = \frac{GM}{r}$$

$$\left(\frac{2\pi r}{T}\right)^2 = \frac{GM}{r^2}$$

$$M = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2}{G} \left(\frac{r^3}{T^2}\right)$$

Here $r = 1.49 \times 10^{11} \text{ m}$,

$G = 6.66 \times 10^{-11} \text{ SI units}$

$T = 365 \times 86400 \text{ s}$.

$$M = \left[\frac{4 \times (3.142)^2}{6.66 \times 10^{-11}} \right] \left[\frac{(1.49 \times 10^{11})^3}{(365 \times 86400)^2} \right]$$

$$M = 1.972 \times 10^{30} \text{ kg}$$

Example 5.20. An artificial satellite is circling round the earth at a distance of 900 km from the surface. Calculate its velocity and period of revolution. Radius of the earth = 6400 km and $g = 9.8 \text{ m/s}^2$. [Delhi, 1976]

Let the radius of the orbit be R and radius of the earth r .

$$r = 64 \times 10^5 \text{ m}$$

$$R = 6.4 \times 10^6 + 9 \times 10^5$$

$\therefore$

$$R = 7.3 \times 10^6 \text{ m}$$

$$\frac{g'}{g} = \frac{r^2}{R^2}$$

$$g' = \frac{gr^2}{R^2}$$

Also

$$g' = \frac{v^2}{R}$$

$$\frac{v^2}{R} = \frac{gr^2}{R^2}$$

$$v^2 = \frac{gr^2}{R}$$

$$v = \left[\frac{9.8 \times (6.4 \times 10^6)^2}{7.3 \times 10^6} \right]^{\frac{1}{2}}$$

$$v = 7.415 \times 10^3 \text{ m/s} = 7.415 \text{ km/s}.$$

$$T = \frac{2\pi R}{v} = \frac{2 \times 3.142 \times 7.3 \times 10^6}{7.415 \times 10^3}$$

$$T = 6186 \text{ s}$$

$$T = 1 \text{ hr} - 43 \text{ min} - 6 \text{ s}$$

Example 5.21. Prove that the least velocity with which a particle must be projected from the surface of a planet of radius R and density ρ in order that it may escape completely is

(Delhi, 1976)

$$\sqrt{\frac{8\pi G \rho R^2}{3}}$$

The escape velocity is given by,

$$v = \sqrt{2gR}$$

But

$$g = \frac{GM}{R^2} = \frac{G\left(\frac{4}{3}\pi R^3\right)\rho}{R^2}$$

$$g = G\left(\frac{4}{3}\pi R\right)\rho$$

$$\therefore v = \sqrt{2G\left(\frac{4}{3}\pi R\right)\rho R} = \sqrt{\frac{8\pi G\rho R^3}{3}}$$

Example 5.22. Calculate the limiting velocity required by an artificial satellite for orbiting round the earth. Given that the radius of the earth = 6.4×10^6 m and $g = 9.8$ m/s².

(Gorakhpur, 1977)

Here

$$v = \sqrt{gR}$$

$$g = 9.8 \text{ m/s}^2, \quad R = 6.4 \times 10^6 \text{ m}$$

$\therefore$

$$v = \sqrt{9.8 \times 6.4 \times 10^6}$$

$$v = 7.92 \times 10^3 \text{ m/s}$$

$$v = 7.92 \text{ km/s}$$

Example 5.23. If a body of mass 10 kg is at rest at a distance r above the surface of the earth, what velocity should be given to it to enable it to escape from the earth's gravitational field?

$$G = 6.67 \times 10^{-11} \text{ SI units}, \quad M = 5.96 \times 10^{24} \text{ kg.}$$

$$r = 6.4 \times 10^6 \text{ m}$$

(Lucknow, 1976)

Let the distance of the body from the centre of the earth be R .

$$R = 2r = 2 \times 6.4 \times 10^6 = 12.8 \times 10^6 \text{ m}$$

Suppose escape velocity is v

$$\therefore \int_R^{\infty} \left(\frac{GMm}{R^2} \right) dR = \frac{1}{2} mv^2$$

$$v = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.96 \times 10^{24}}{12.8 \times 10^6}}$$

$$v = 7.92 \times 10^3 \text{ m/s}$$

$$v = 7.92 \text{ km/s}$$

Example 5.24. Calculate the escape velocity of a body from the moon's surface, if the radius of the moon is 1.7×10^6 m and acceleration due to gravity on the moon's surface is 1.63 m/s².

(Bombay, 1971)

Here

$$r = 1.7 \times 10^6 \text{ m}$$

$$g = 1.63 \text{ m/s}^2$$

$$v = \sqrt{2gr} = \sqrt{2 \times 1.63 \times 1.7 \times 10^4}$$

$$v = 2.354 \times 10^3 \text{ m/s}$$

$$v = 2.354 \text{ km/s}$$

Example 5.25. Moon describes a circular orbit of radius 4×10^8 m around the earth in 28 days. Titan describes a circular orbit of radius 1.6×10^9 m around the saturn in 16 days. Compare the masses of saturn and the earth. (Bombay 1974)

Let the mass of the earth be M_1 and mass of saturn M_2

For the earth,

$$M_1 = \frac{4\pi^2}{g} \left(\frac{r_1^3}{T_1^2} \right) \quad \dots(i)$$

For the saturn

$$M_2 = \frac{4\pi^2}{g} \left(\frac{r_2^3}{T_2^2} \right) \quad \dots(ii)$$

Divide (ii) by (i)

$$\frac{M_2}{M_1} = \left(\frac{r_2}{r_1} \right)^3 \left(\frac{T_1}{T_2} \right)^2$$

Here

$$r_1 = 4 \times 10^8 \text{ m}$$

$$r_2 = 1.6 \times 10^9 \text{ m}$$

$$T_1 = 28 \times 86400 \text{ s}$$

$$T_2 = 16 \times 6400 \text{ s}$$

$$\therefore \frac{M_2}{M_1} = \left(\frac{1.6 \times 10^9}{4 \times 10^8} \right)^3 \left(\frac{28 \times 86400}{16 \times 86400} \right)^2$$

$$\frac{M_2}{M_1} = 196$$

Example 5.26. What will be the acceleration due to gravity on the surface of the moon if its radius were $\frac{1}{4}$ th radius of the earth and mass $\frac{1}{80}$ th of the mass of the earth? What will be the escape velocity on the surface of moon if it is 11.2 km s^{-1} on the surface of the earth?

[Roorke Engineering Entrance Exam., 1983]

$$g = \frac{GM}{R^2} \quad \dots(i)$$

$$g' = \frac{GM'}{(R')^2} \quad \dots(ii)$$

$$\therefore \frac{g'}{g} = \left(\frac{M'}{M} \right) \left(\frac{R}{R'} \right)^2$$

$$\text{Here } \frac{M'}{M} = \frac{1}{80} \text{ and } \frac{R'}{R} = \frac{1}{4}, g = 9.8 \text{ m/s}^2$$

$$\therefore g' = 9.8 \left(\frac{1}{80} \right) \left(4 \right)^2$$

Escape velocity on the surface of the earth,

$$v = \sqrt{2gR} \quad \dots(iii)$$

Escape velocity on the surface of the moon

$$v' = \sqrt{2g'R'} \quad (iv)$$

$$\therefore \frac{v'}{v} = \sqrt{\left(\frac{g'}{g}\right)\left(\frac{R'}{R}\right)}$$

$$v' = v \sqrt{\left(\frac{g'}{g}\right)\left(\frac{R'}{R}\right)}$$

$$v' = 11.2 \left[\left(\frac{1.96}{9.8} \right) \left(\frac{1}{4} \right) \right]^{\frac{1}{2}}$$

$$v' = 2.504 \text{ km/s.}$$

5.32. Gravitational Self Energy

A body has gravitational self energy due to its mass. It is the potential energy of the body due to its mass. It is equal to the work done in bringing infinite small masses at infinity to a point where the body is formed. As two masses experience force of attractions therefore the self energy will be negative.

Suppose a body consists of N particles each of mass $m_1, m_2, m_3, m_4, \dots$ etc., then

$$U = -\frac{1}{2} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}}$$

Negative sign indicates that the energy is released when masses are combined to form a bigger mass.

5.33. Gravitation Self Energy of a Sphere

Consider a body of mass M , radius R and density ρ . Consider an element of a body of radius r and density ρ .

The mass of the element $= \frac{4}{3}\pi r^3 \rho$.

Now consider that at the stage of formation, an additional mass is given to the element so that its radius $= (r + dr)$.

Therefore additional mass $= 4\pi r^2 (dr) \rho$.

The energy released in increasing the mass is equal to the work done in bringing this mass from infinity to the sphere.

$$\therefore dU_s = - \frac{G \left(\frac{4}{3}\pi r^3 \rho \right) [4\pi r^2 (dr) \rho]}{r}$$

$$dU_s = - \frac{16\pi^2 G \rho^2 r^4 (dr)}{3} \quad \dots(i)$$

Therefore total energy released in the formation of the whole of the body of radius R will give the value of the self energy the sphere.

Therefore gravitational self energy of the sphere.

$$U_s = \int dU_s = - \int_0^R \left(\frac{16\pi^2 \rho^2 G}{3} r^4 dr \right)$$

$$U_s = - \left(\frac{16}{15} \right) \pi^2 \rho^2 G R^5 \quad \dots(ii)$$

But

$$\frac{4}{3} \pi R^3 \rho = M$$

$$\therefore U_s = - \left(\frac{3}{5} \right) \left(\frac{GM^2}{R} \right) \quad \dots(iii)$$

Gravitational Self Energy of the Sun

In the case of the sun,

$$U_s = - \frac{3}{5} \left(\frac{GM^2}{R} \right)$$

Here M is the mass of the sun and R is the radius of the sun.Taking mass of the sun, $M = 2 \times 10^{30}$ kgradius of the sun, $R = 7 \times 10^8$ m

$$G = 6.67 \times 10^{-11} \text{ SI units,}$$

The gravitational self energy of the sun,

$$U_s = - \frac{3 GM^2}{5 R}$$

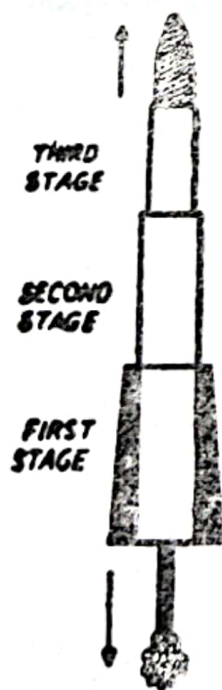
$$U_s = - \frac{3 \times 6.67 \times 10^{-11} \times (2 \times 10^{30})^2}{5 \times 7 \times 10^8}$$

$$U_s = - 2.2868 \times 10^{11} \text{ J}$$

Note. When a planet or a body contracts without change in mass (i.e., its density increases), energy will be released.

5.34. Rocket

A rocket is based on the principle of conservation of momentum. It has a combustion chamber in which the fuel (solid or liquid) is burnt. The burnt gas escaping in a vertically downward direction gives an equal vertical upward momentum to the rocket (Fig. 5.28).



Suppose the initial mass of the rocket is M and the upward velocity is V . Let, in a short time t , the mass of the gas escaping be m . The velocity of the escaping gas relative to the rocket $= V_r$. Therefore the velocity of the gas with respect to the earth,

$$V_1 = V - V_r$$

The momentum of the escaping gas

$$mV_1 = m(V - V_r) \text{ downwards.}$$

Due to this, the rocket gains a momentum

$$= m(V - V_r) \text{ upwards}$$

Initial momentum of the rocket $= MV$ If the increase in velocity of the rocket is v , the momentum of the rocket

$$= (M - m)(V + v)$$

Total change in momentum of the system

$$= (M - m)(V + v) + m(V - V_r) - MV$$

But $F \times t = \text{change in momentum}$

Fig. 5.28

$$\therefore F \times t = (M - m)(V + v) + m(V - V_r) - MV$$

Here $F = -Mg$

(The negative sign shows that the weight acts in the downward direction and the force in the upward direction).

$$\therefore -Mgt = (M-m)(V+v) + m(V-V_r) - MV$$

Expanding and neglecting mv , we get

$$-Mgt = Mv - mV_r$$

$$\frac{Mv}{t} = V_r \left(\frac{m}{t} \right) - Mg \quad \dots(i)$$

Here, acceleration, $a = \frac{v}{t}$

$$\therefore Ma = V_r \left(\frac{m}{t} \right) - Mg$$

or
$$a = \left(\frac{V_r}{M} \times \frac{m}{t} \right) - g \quad \dots(ii)$$

The term $\left(\frac{V_r}{M} \times \frac{m}{t} \right)$ is the upward acceleration of the rocket

and g is the downward acceleration due to gravity. As the rocket ascends higher, the value of g decreases and consequently a increases. Also due to the decrease in the value of M , the acceleration of the rocket increases. Thus the acceleration of the rocket continuously increases till the whole of the fuel is burnt.

In practice multi-stage rockets are used. As soon as the fuel of the first stage is consumed, that stage separates from the main rocket.

5.35. Jet Plane

The principle of working of a jet plane is similar to that of a rocket. A jet plane is fitted with a diesel engine. The air is sucked in from the front and the fuel is sprayed into it. This mixture is compressed to a high pressure in the piston chamber and ignition takes place. The burnt gases escape with a high velocity from the rear and impart an equal and opposite momentum to the jet plane in accordance with Newton's third law of motion. These planes can fly at high speeds. If the speed is greater than the velocity of sound, they are called supersonic jets.

The difference between a rocket and a jet plane is that, in the case of a rocket the fuel burns quickly and imparts large thrust. In the case of a plane the consumption of the fuel is continuous.

Example 5.27. In the first second of its flight a rocket ejects fuel equal to $\frac{1}{50}$ th of its mass with a relative velocity of 2500 m/s. Calculate the acceleration of the rocket. ($g = 9.80 \text{ m/s}^2$)

$$a = \left(\frac{V_r}{M} \times \frac{m}{t} \right) - g$$

Here $V_r = 2500 \text{ m/s}$, $\frac{m}{M} = \frac{1}{50}$, $g = 9.80 \text{ m/s}^2$

$$t = 1 \text{ s}$$

$$a = \left(\frac{2500 \times 1}{50} \right) - 9.80 = 40.20 \text{ m/s}^2$$

Example 5-28. A balloon has a total mass of 300 kg including the air contained in it and a volume of 300 m³. Calculate the resultant upward force on the balloon. Density of air = 1.20 kg/m³. and 1 kg wt = 10 N.

Assuming that the upward force remains constant for 10 s, calculate

- (i) the upward acceleration of the balloon,
 (ii) the velocity attained after 10 s and
 and (iii) the change in height during these 10 s.

Weight of the balloon = 300 kg wt

$$= 300 \times 10 = 3 \times 10^3 \text{ N}$$

Upward thrust

$$= 300 \times 1.20 \text{ kg wt}$$

$$= 300 \times 1.20 \times 10 \text{ N}$$

$$= 3.6 \times 10^3 \text{ N}$$

Resultant upward force = $3.6 \times 10^3 - 3 \times 10^3$

$$= 600 \text{ N}$$

∴ (i)

(i) Force = mass × acceleration

$$\therefore \text{Acceleration} = \frac{\text{Force}}{\text{Mass}}$$

$$a = \frac{600}{300} = 2 \text{ m/s}^2 \quad \dots (ii)$$

(ii) $v = at$

$$v = 2 \times 10 = 20 \text{ m/s upwards} \quad \dots (iii)$$

(iii) $h = \frac{1}{2} at^2$

$$h = \frac{1}{2} \times 2 \times (10)^2$$

$$h = 100 \text{ m} \quad \dots (iv)$$

Exercises 5

1. State and explain Kepler's laws of planetary motion.
2. Show that the force of attraction between two bodies is inversely proportional to the square of the distance between them.
3. State Kepler's laws of planetary motion. Show that the law of areas is merely a statement of the law of conservation of angular momentum. [Delhi Engineering, 1980]
4. Write down Kepler's laws of planetary motion. Prove that the first and the second laws lead to the conservation of angular momentum vector. [Calcutta (Hons) 1980]
5. State Newton's universal law of gravitation. Give the dimensions and S.I. units of gravitational constant.

(Delhi 1978)

6. Describe Cavendish method for determining the value of gravitational constant. Mention the drawbacks and the sources of error of the experiment. (Delhi 1979)
7. Describe Boy's method for determining G . What are the merits of this method over Cavendish's method. (Delhi 1976, 78, 80, Mysore 1977)
8. Describe briefly Poynting method for determining the gravitational constant.
9. State Newton's law of gravitation. Describe a method of determining the value of the constant of gravitation. How can the values of the mass and the density of the earth be found out, if the value of the above constant is known? [Calcutta, 1979]
10. Derive an expression for the time period of oscillation of a compound pendulum. [Delhi (Sub) 1978]
Delhi (Hons) 1978
(Mysore 1978)
11. Give the theory of Kater's reversible pendulum and explain how acceleration due to gravity is determined with it. [Delhi (Sub) 1977, 80]
12. Discuss the variation of acceleration due to gravity with (1) altitude (2) depth and (3) rotation of the earth.
13. Show that the object lying at the equator will fly off the surface of the earth, if the speed of rotation of the earth increases to approximately seventeen times its present value.
14. Distinguish between the following :—
(a) mass and weight
(b) gravity and gravitation
(c) inertial mass and gravitational mass.
15. Explain the terms (i) gravitational field (ii) gravitational potential and (iii) gravitational potential energy.
16. Calculate the gravitational potential and field due a sphere at a point (i) outside the shell (ii) inside the shell. (Delhi 1978)
17. Obtain the expressions for gravitational potential and field due to a uniform solid sphere at a point (i) inside the sphere and (ii) outside the sphere. (Delhi (Sub) 1976, 1980)
18. Calculate the gravitational potential and gravitational field due to a hollow sphere (i) at a point outside the sphere, (ii) at a point inside the hollow space and (iii) at a point inside the material. [Kanpur, 1981]
19. Show that the gravitational field due to thin uniform circular plate of radius a at point distant R from the centre and on the axis passing through the centre and perpendicular to the plane of the plate is given by

$$E = -2\pi G \rho \left[1 - \frac{R}{(a^2 + R^2)^{1/2}} \right]$$

20. What is an artificial satellite? Show that the orbital velocity is given by

$$v = G^{1/2} M^{1/2} r^{-1/2}$$

21. What is a stationary satellite? Calculate the height of a stationary satellite coplanar with the earth's equator.
[Delhi (Sub) 1980]

22. Define escape velocity. Show that the escape velocity from the surface of the earth is equal to 11.2 km/s.

23. Distinguish between orbital velocity and escape velocity. What are their values near the earth's surface? (Delhi 1979)

24. Describe an accurate method for determining the value of G .
[Delhi (Sub) 1976]

24. Prove that the least velocity with which a particle must be projected from the surface of the radius R and density ρ in order that it may escape completely is

$$R \left[\frac{8\pi G \rho}{3} \right]^{1/2}$$

[Delhi (Sub) 1979]

26. Describe an accurate method for the measurement of g . How does g vary with (a) altitude (b) depth?

[Delhi (Sub) 1978, 80]

27. Define g and G . Do you expect the values of these to vary from place to place? Give reasons.
[Delhi (Sub) 1977]

28. What do you understand by acceleration due to gravity? How does g vary with latitude and altitude? Derive the necessary expression.
[Delhi (Sub) 1980]

29. Find the velocity of projection of a body so that it may (i) revolve round the earth in a circle near its surface and (ii) just escape the gravitational field of the earth.

(Delhi 1979)

[Ans. (i) 7.92 km/s in the horizontal direction
(ii) 11.2 km/s vertically upwards]

30. What is meant by gravitational potential and field intensity at a point in the gravitational field? How are they related?

Calculate these quantities due to a sphere at a point inside a uniform solid sphere.
[Delhi (Sub) 1979]

31. Calculate the limiting velocity required by an earth's satellite for orbiting round the earth. Radius of the earth = 6.4×10^6 m and $g = 9.80$ m/s².
[Delhi (Sub) 1979]

[Ans. 7.92 km/s]

32. Show that due to earth's rotation, the observed value of acceleration due to gravity at a place of latitude ϕ , is given by

$$g_{\phi} = g \left[1 - \frac{r\omega^2 \cos^2 \phi}{g} \right] \quad (\text{Delhi 1979})$$

33. Show that if the earth suddenly starts rotating faster than the present rate such that $\frac{r\omega^2}{g}$ becomes 1,

$$g_{\phi} = g [1 - \cos^2 \phi] = g \sin^2 \phi \quad (\text{Delhi 1979})$$

34. State Kepler's laws. Explain them in case of a body moving around another body in a circular orbit.
35. An artificial earth satellite is placed in a stable circular orbit above the equator such that it remains stationary with respect to an observer on the earth. What is the radius of its orbit?

$$G = 6.67 \times 10^{-11} \text{ S I units}$$

$$\text{Mass of the earth} = 5.98 \times 10^{24} \text{ kg}$$

$$\begin{aligned} & [\text{Calcutta 1978}] \\ & [\text{Ans. } 3.56 \times 10^7 \text{ m}] \end{aligned}$$

36. How much faster than the present speed would the earth have to rotate on its axis so that the apparent weight of bodies at the equator is zero?
[Delhi (Sub) 1980]
[Ans. 17]
37. Define and explain gravitational potential. Derive an expression for it at a point inside and outside a thin uniform spherical shell.
[Delhi (Hons) 1982]
38. Calculate the gravitational self energy of the Sun.
[Mass of Sun = 2×10^{30} kg. Radius of Sun = 7×10^8 m,
 $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.]
[Delhi (Hons) 1982]
[Ans. $-2.2868 \times 10^{41} \text{ J}$]
39. Define and explain the terms 'Gravitational Potential' and 'Gravitational Field'
[Delhi (Sub) 1982]
40. Obtain expressions for the gravitational potential and gravitational field due to a sphere at a point inside the sphere.
[Delhi (Sub) 1982]
41. Define g . How does it vary with (i) altitude and (ii) depth?
[Delhi (Sub) 1982]
42. How far away from the earth does the acceleration due to gravity become 10% of its value on the surface of the earth?
Radius of the earth = 6.38×10^8 m.
[Delhi (Sub) 1982]
43. Prove that the escape velocity from the surface of earth is $\sqrt{2}$ times the velocity of projection of an artificial satellite orbiting close around the earth.
[Delhi, 1982]

44. Obtain the value of escape velocity for an atmospheric particle 1000 km above the surface of the earth.
- Mass of the earth $= 5.98 \times 10^{24}$ kg
 Radius of the earth $= 6.37 \times 10^3$ km
 Gravitational constant $= 6.67 \times 10^{-11}$ N-m²/kg².
[Delhi, 1982]
45. Find the gravitational potential due to a spherical shell at a point outside the shell.
[Lucknow, 1982]
46. Define gravitational potential and calculate the gravitational potential of mass m near the surface of the earth.
[Rohilkhand, 1982]
47. State and prove the Kepler's second law for planetary motion.
[Rohilkhand, 1982]
48. Obtain an expression for the escape velocity of a body from the surface of the earth. Calculate its value. [Rohilkhand, 1982]
49. Explain what is meant by gravitational field and gravitational potential. Calculate the gravitational potential and field at a point (i) inside and (ii) outside a hollow spherical shell.
[Delhi, 1982]
50. A satellite is in a circular orbit near a planet's surface and its orbital speed is v_0 . Show that the escape velocity for any object from that planet is $v_0\sqrt{2}$. [I.A.S., 1982]
51. For a satellites in circular orbit around a planet show that the square of the period is proportional to the cube of the radius of the orbit. [I.A.S., 1982]
52. Compute the gravitational self energy of a solid spherical body.
[Delhi (Hons) 1984]
53. Write down the Law of Universal Gravitation in vector notation. Deduce Kepler's laws in the case of motion of a planet around the sun taking its mass to be finite.
[Delhi (Hons) 1985]